

Provisional figures and tables

MTU15 thesis (companion to paper.tex)

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This file is exploratory scratch space. None of these objects is currently cited from `paper.tex`. They are kept here for inspection and decision-making — promote to the main paper if the empirical content is load-bearing, otherwise leave here or move to `attic/`.

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1 Reform-window identification (DA15 + IDA15)

The headline thesis question is the effect of **MTU15 granularity** on strategic bidding. Re-forzada (the post-blackout RT2 pay-as-bid + zonal-monopoly windfall channel documented in §7–§8–§11) is a parallel *confounding* mechanism — analogous in spirit to the operational/solar midday channel in the main paper (`paper.tex` §2.4): a real economic phenomenon that we want to characterise but *exclude from the identifying contrast* so as not to mix channels. This section’s design picks reform windows in which reforzada is held constant, then differences out residual seasonality via a 2024 placebo.

Design. Two reform-windows, each chosen so that reforzada is constant within both the pre- and post-treatment sub-windows:

Table 1: Reform-window design. Reforzada is the post-blackout RT2 pay-as-bid regime (active 2025-04-28 onward). The 2024 placebo uses calendar-matched dates. PT assumption: YoY seasonal change in (crit – flat) was the same in 2024 as in 2025.

Reform / sample	Pre-window	Post-window	Reforzada	Length pre/post
IDA15 (2025)	2024-12-09 – 2025-03-18	2025-03-19 – 2025-04-27	off	100 / 40 d
IDA15 placebo (2024)	2023-12-09 – 2024-03-18	2024-03-19 – 2024-04-27	off	matching
DA15 (2025)	2025-04-28 – 2025-09-30	2025-10-01 – 2026-02-13	on	155 / 135 d
DA15 placebo (2024)	2024-04-28 – 2024-09-30	2024-10-01 – 2025-02-13	off	matching

Reading (supportive, by design). DA15 holds reforzada constant (on) in both sub-windows; IDA15 holds it constant (off). Each pre-window is ≥ 3 months; DA15 has 5 months post-reform.

Verdict (supportive). Temporal design isolates each MTU15 reform from reforzada. The 2024 placebo provides the seasonality control absent on the within-day DiD alone.

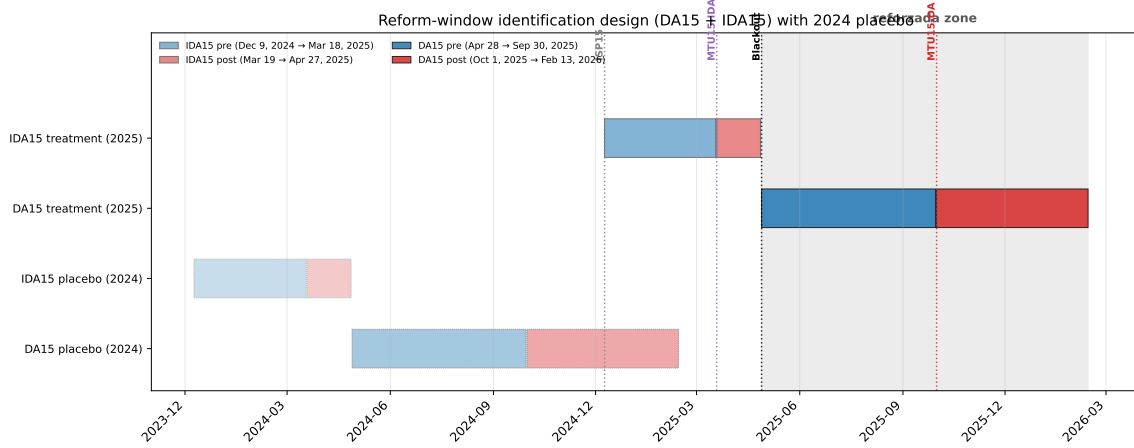


Figure 1: Timeline of reforms and DDD windows. Vertical guides: ISP15 (2024-12-09), MTU15-IDA (2025-03-19), Iberian blackout (2025-04-28), MTU15-DA (2025-10-01). Shaded grey: reforzada zone. Blue bars: pre-reform sub-windows; red bars: post-reform. 2024 placebo rows shown with lighter shading.

Reading (supportive). Visual confirmation: DA15 windows sit entirely inside the reforzada zone; IDA15 windows entirely outside.

Triple-difference specification. Let $Y_{f,u,d,h}$ be the outcome at firm f , unit u , date d , clock-hour h . Define $\text{crit}_h = 1$ if $h \in \{5-9, 16-22\}$ (treated hour-class), restrict sample to $\{\text{crit}, \text{flat}\}$. Let $\text{post}_d = 1$ if d is after the reform date, $y25_d = 1$ if d falls in the 2025 (treatment) window. Then:

$$Y_{f,u,d,h} = \alpha + \beta_1 \text{crit}_h + \beta_2 \text{post}_d + \beta_3 y25_d + \beta_4 (\text{crit} \times \text{post}) + \beta_5 (\text{crit} \times y25) + \beta_6 (\text{post} \times y25) + \boxed{\beta_7 (\text{crit} \times \text{post} \times y25)} + \gamma_f + \mu_u + \delta_{\text{DOW}(d)} + \varepsilon_{f,u,d,h}. \quad (1)$$

β_7 is the headline coefficient: the change in the (critical – flat) differential at the reform date, *net of* the corresponding 2024 seasonal change. FE: firm γ_f + unit μ_u + day-of-week δ_{DOW} . SE clustered at the date level. Equivalently $\beta_7 = \Delta\Delta Y^{2025} - \Delta\Delta Y^{2024}$ where $\Delta\Delta Y = (\bar{Y}_{\text{post,crit}} - \bar{Y}_{\text{post,flat}}) - (\bar{Y}_{\text{pre,crit}} - \bar{Y}_{\text{pre,flat}})$.

Event-study spec (visual pre-trends). For each outcome, plot the monthly differential separately for the 2025 treatment year and the 2024 placebo year, anchored to $\tau =$ months from the reform date:

$$\Delta Y_{\tau,y} = \bar{Y}_{\text{crit}}(\tau, y) - \bar{Y}_{\text{flat}}(\tau, y), \quad y \in \{2024, 2025\}.$$

Identifying assumption test: for $\tau < 0$ the two series should overlap (parallel pre-trends); at $\tau \geq 0$ the 2025 series diverges from the 2024 series by exactly the treatment effect.

Identifying assumptions. *Formal PT (triple-difference).*

$$\mathbb{E}[\Delta\Delta Y(0) \mid y25 = 1] = \mathbb{E}[\Delta\Delta Y(0) \mid y25 = 0],$$

where $\Delta\Delta Y(0)$ is the untreated within-day pre/post change in the critical-flat differential.

Threats killed by the design.

- *Seasonality of the critical-flat spread* (summer-to-winter heating/AC patterns): differenced out by the 2024 placebo on the same calendar months.
- *Day-level cost shocks* (TTF gas, CO₂, demand spikes): within-day differencing on each date absorbs anything that hits both hour-classes proportionally.
- *Asymmetric gas-cost confound*: ruled out empirically in `paper.tex` §2.4 — CCGT price-setting share 9.1% crit vs 9.0% flat, so TTF shocks cannot propagate asymmetrically.
- *Operational midday channel* (solar drives prices down at midday): midday hours $\in \{11-14\}$ excluded from the control set; only flat hours $\in \{1, 2, 3\}$ enter as control.
- *Reforzada / RT2 pay-as-bid windfall*: held constant within each window by design (the central insight). Characterised as confounder in §7–§8–§11.

Threats remaining.

- *Within-firm portfolio composition shifts*: addressed by unit FE; sensitivity check restricts to units active in both years.
- *Reform-anticipation effects*: tested by the event-study leads β_τ for $\tau \in \{-1, -2\}$.

Extension to k -way DiD (pivotal moderator). The main paper’s appendix already runs a triple-difference with $\text{pivotal}_f \in \{0, 1\}$ (Big-4 dominant firms vs non-pivotal placebos). Adding the 2024 placebo as a fourth axis gives a **quadruple difference**. The estimable spec, after dropping the pivotal_f main effect (firm-invariant, absorbed by γ_f):

$$\begin{aligned} Y_{f,u,d,h} = & \alpha + \beta_1 \text{crit} + \beta_2 \text{post} + \beta_3 y25 + \sum_{\substack{6 \text{ pairs involving crit, post, y25, pivotal}}} \\ + & \sum_{\substack{4 \text{ triples involving time-varying axes}}} + \boxed{\beta_{1234} (\text{crit} \times \text{post} \times y25 \times \text{pivotal})} + \gamma_f + \mu_u + \delta_{\text{DOW}(d)} + \varepsilon. \end{aligned} \tag{2}$$

The pivotal main effect and any $\text{pivotal} \times (\text{firm-invariant})$ interaction are dropped — absorbed by firm FE γ_f . All 6 pairs and all 4 triples that involve at least one within-firm-varying axis (crit, post, y25) are kept.

Sample restriction for the DDDD. To avoid contamination from the June 2024 IDA reform (6→3 sessions) and the December 2024 ISP15 rollout, the DDDD sample is restricted to **Jan–May 2024 (placebo) vs Jan–May 2025 (treatment)**. The 2-month post-IDA15 window

(Mar 19 – Apr 27 2025) sits inside this restriction; the 2024 placebo correspondingly uses Mar 19 – Apr 27 2024.

SE clustering. Cluster at the **firm** level, not the unit level: pivotal does not vary within firm, so within-firm error correlation is the relevant dependence.

Interpretation. β_{1234} identifies the *pivotal-differential* MTU15 effect on the critical-flat differential — whether pivotal firms respond more strongly than non-pivotal firms, net of all lower-order interactions. It is a direct test of the *market-power mechanism*: a significant $\beta_{1234} > 0$ (in the strategic-exploitation direction) or < 0 (operational direction) would indicate the effect is concentrated in the firms with pivotality. **Non-rejection of $\beta_{1234} = 0$ does *not* rule out a uniform-effect mechanism** in which all firms respond similarly to the reform — which would still show up in β_7 . The two coefficients answer different questions: β_7 is the average reform effect; β_{1234} is the pivotal-incremental effect. We will report both with their CIs side-by-side so the power gap (the DDDD necessarily has wider SE because it conditions on a higher-order interaction) is visible.

Outcomes. Three outcomes, each with its own identifying assumption and economic story. $D_w/\Delta p/\Delta q$ are *not* DDD outcomes — they require within-hour quarters that do not exist pre-MTU15 (see §10, §12 for the post-only cross-section).

Table 2: DDD outcomes, with competing theoretical predictions. The headline coefficient is β_7 on (crit $\times$ post $\times$ y25). Sign predictions are *not* known a priori — both directions are theoretically plausible.

Outcome	Theory pulling $\beta_7 < 0$	Theory pulling $\beta_7 > 0$
DA, IDA clearing price	Granularity lets dispatchable units re-price into critical quarter-slots from solar/wind margin: peak-flat spread compresses.	Granularity gives strategic firms within-hour shaping leverage; firms exploit the new dimension to push critical-quarter prices up.
q_1 (DA-cleared MWh / unit-hour)	RT2 backstop $\Rightarrow$ firms bid above clearing in critical hours and wait for IDA/RT rescue: DA volume drops in crit.	Granularity gives operational dispatchable units new clearing opportunities in critical hours; DA volume rises.
q_2 (intraday upward MWh / unit-hour)	Operational: granularity reduces forecast-error from coarse hourly programs $\Rightarrow$ less repositioning needed.	Ito-Reguant: firms withhold in DA, push back via IDA $\Rightarrow$ more repositioning.

Reading (pending, sign of β_7 uncertain ex ante). For all three outcomes, theory admits competing predictions. The empirical sign tells us which mechanism dominates: operational refinement (negative for prices, positive for q_1 , negative for q_2) vs strategic exploitation (positive for prices, negative for q_1 , positive for q_2). The DDD provides the directional test.

How to read Figure 2. Each panel is a single (outcome $\times$ reform) cell. The horizontal axis is τ = months relative to the reform date (e.g. $\tau = 0$ is March 2025 for IDA15 and October 2025 for DA15). The vertical axis is the monthly mean of the within-day differential $\bar{Y}_{\text{crit}}(\tau) - \bar{Y}_{\text{flat}}(\tau)$ in that month — a scalar that tells us how much higher the critical-hour outcome is relative to the same day’s flat-hour outcome.

Each panel has two coloured lines:

- **Red (2025):** the actual treatment year. At $\tau = 0$, MTU15 is implemented.
- **Blue (2024):** the calendar-matched placebo. At $\tau = 0$, the analogous calendar month one year earlier — no reform, just seasonality.

The DDD coefficient β_7 is a difference of *changes* (not levels). In raw form the two lines can sit at different levels (e.g., 2025 prices systematically higher than 2024 because of inflation) and the design still works — the level shift is absorbed by the y25 main effect β_3 and what matters is that the (red – blue) gap stays constant pre-reform and shifts post-reform.

The figure normalises each line by subtracting its value at $\tau = -1$, so both start at zero at the omitted reference month. After normalisation the visual test is the standard event-study form:

- For $\tau < 0$ (pre-reform): red and blue should **overlap** (parallel-trends test).
- At $\tau \geq 0$ (post-reform): a persistent vertical separation between red and blue is the treatment effect.
- β_7 is the average post-period vertical separation between the two normalised lines.

Event-study: monthly (critical – flat) differential, 2025 treatment vs 2024 placebo. Each line normalised to 0 at $\tau = -1$ (subtraction of the $\tau = -1$ value). Pre-trend test (textit{parallel slopes}): for $\tau < 0$, the two lines should track each other. Treatment effect at $\tau \geq 0$: the vertical gap between the red and blue lines.

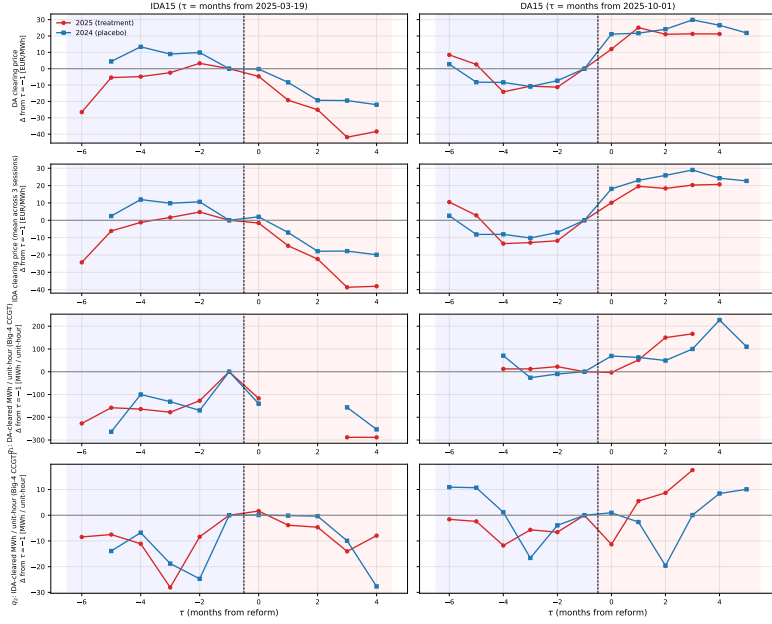


Figure 2: Event-study (reduced-form, normalised). Rows = outcomes (DA price, IDA price, q_1 , q_2); columns = reforms (IDA15, DA15). Each panel plots the monthly (critical – flat) within-day differential, anchored to $\tau =$ months from the reform date and *normalised by subtracting the $\tau = -1$ value*, so both lines start at 0 at $\tau = -1$. Red = 2025 (treatment); blue = 2024 (placebo). Pre/post regions lightly shaded. Reduced-form monthly means; full regression-based event-study with CI follows the headline DDD.

Reading (mixed, visual). DA15 column (right): DA price and IDA price show modest 2025 vs 2024 separation at $\tau \geq 0$; q_1 shows the 2025 series sitting below the 2024 series *already before* $\tau = 0$ — the reforzada channel pre-dates MTU15-DA and pre-bends the differential, making the parallel-trends test fail for q_1 on this window; q_2 noisy. IDA15 column (left): pre-period overlap looks reasonable for prices, post-period too short to read. Pre-regression visual only; the headline DDD with date-clustered SE follows.

Verdict (pending). Headline DDD with SE clustered at the date level required to confirm. The visual is supportive for prices on DA15; for volumes the reforzada pre-trend complicates the read.

Estimators in practice: OLS-FE and doubly-robust DiD on prices in levels. We report two estimators for the same DDD parameter, each with three to four specifications. Both use clearing prices in EUR/MWh as the outcome (in levels), and both target the same coefficient: the (critical – flat) differential response to the reform, net of the 2024 calendar placebo.

Panel construction. For each (reform window, market) combination $w \in \{\text{DA15, IDA15}\} \times \{\text{DA, IDA-S1}\}$ we build two analytic panels:

- *Period panel*: one row per (date, ISP-period) tuple. Pre-MTU15 days contribute 1 observation per clock-hour (MTU60); post-MTU15 days contribute 4 (MTU15). Sample restricted to clock-hours in $\{\text{critical} \cup \text{flat}\}$. Used for the OLS-FE estimator. Sizes: 13 888 obs (DA15-DA), 19 138 obs (DA15-IDA), 3 934 obs (IDA15-DA), 5 530 obs (IDA15-IDA).
- *Spread panel*: one row per date with the within-day means $\bar{Y}_d^{\text{crit}} = \frac{1}{|H_{\text{crit}}|} \sum_{p \in H_{\text{crit}}} Y_{d,p}$ and $\bar{Y}_d^{\text{flat}}$ as two columns. Used for the DR-DiD estimator (one drdid call per hour-class). Sizes: 584 dates (DA15-DA), 554 (DA15-IDA), 281 (IDA15-DA), 278 (IDA15-IDA).

OLS-FE DDD. On the period panel, three specifications, each with the same DDD coefficient β_7 but different fixed-effect structures:

$$\begin{aligned}
\text{(M1) sparse: } Y_{d,p} &= \alpha + \sum_{k=1}^7 \beta_k Z_k + \gamma_h + \delta_{\text{DOW}(d)} + \varepsilon_{d,p}, \\
\text{(M2) month FE: } Y_{d,p} &= \alpha + \sum_{k=1}^7 \beta_k Z_k + \gamma_h + \delta_{\text{DOW}(d)} + \eta_{\text{month}(d)} + \varepsilon_{d,p}, \\
\text{(M3) date FE: } Y_{d,p} &= \alpha + \beta_1 \text{crit} + \beta_4 (\text{crit} \times \text{post}) + \beta_5 (\text{crit} \times \text{y25}) \\
&\quad + \beta_7 (\text{crit} \times \text{post} \times \text{y25}) + \gamma_h + \delta_{\text{DOW}(d)} + \nu_d + \varepsilon_{d,p}.
\end{aligned}$$

Here $(Z_1, \dots, Z_7) = (\text{crit}, \text{post}, \text{y25}, \text{crit} \times \text{post}, \text{crit} \times \text{y25}, \text{post} \times \text{y25}, \text{crit} \times \text{post} \times \text{y25})$, γ_h is a clock-hour FE, $\delta_{\text{DOW}(d)}$ is a day-of-week FE, $\eta_{\text{month}(d)}$ is a calendar-month FE, and ν_d is a date FE that absorbs all date-level terms in M3 (so the date-level $\beta_2, \beta_3, \beta_6$ drop out and β_7 is identified purely from within-date variation). SE clustered at the date level via `reghdfe ... , vce(cluster date)`.

Doubly-robust DiD (Sant'Anna-Zhao 2020). On the spread panel, two 2×2 doubly-robust DiDs per (window, market), one on each hour-class outcome:

$$\widehat{\text{ATT}}_c^{\text{DR}} = \mathbb{E}_n \left[\left(w_1(D) - w_0(D, X; \hat{p}) \right) \left((Y_{\text{post}}^c - Y_{\text{pre}}^c) - (\hat{m}_{\text{post}}^0(X) - \hat{m}_{\text{pre}}^0(X)) \right) \right],$$

with $c \in \{\text{crit}, \text{flat}\}$, treatment indicator $D = \text{y25}_d$, time indicator $t = \text{post}_d$, covariates $X = (\text{i.dow}, \text{i.month})$, and

$$w_1(D) = \frac{D}{\mathbb{E}_n[D]}, \quad w_0(D, X; \hat{p}) = \frac{\hat{p}(X)(1-D)/(1-\hat{p}(X))}{\mathbb{E}_n[\hat{p}(X)(1-D)/(1-\hat{p}(X))]}$$

The default `drdid` estimator (`drimp`) uses inverse-probability-tilting for $\hat{p}(X)$ and weighted least squares for the outcome model $\hat{m}^0(X)$. Three nuisance-model specifications are reported:

- *DR1*: no covariates ($X = \emptyset$). Reduces to the unconditional 2×2 DiD on the spread Y^c .
- *DR2*: $X = (\text{i.dow}, \text{i.month})$, asymptotic SE via the influence function.
- *DR3*: $X = (\text{i.dow}, \text{i.month})$, wild-bootstrap inference (999 Mammen draws, `drdid ... , wboot rseed(4242)`).

DDD via the difference of two doubly-robust DiDs (the headline). The triple difference is the difference of the two hour-class ATTs:

$$\widehat{\text{DDD}} = \widehat{\text{ATT}}_{\text{crit}}^{\text{DR}} - \widehat{\text{ATT}}_{\text{flat}}^{\text{DR}}.$$

Joint inference uses the recentered influence functions $\hat{\psi}_i^c$ saved by `drdid ... , stub(c)`, which satisfy $\sqrt{N}(\widehat{\text{ATT}}^c - \text{ATT}^c) = N^{-1/2} \sum_i \hat{\psi}_i^c + o_p(1)$ (Sant'Anna-Zhao 2020, Theorem 1). Because

the crit and flat means are computed on the same dates (within-date dependence), the variance of the DDD is taken from the variance of the IF *difference*:

$$\widehat{\text{Var}}(\widehat{\text{DDD}}) = \frac{1}{N} \widehat{\text{Var}}_n [\hat{\psi}_i^{\text{crit}} - \hat{\psi}_i^{\text{flat}}], \quad N = \text{number of dates},$$

i.e. the cross-class covariance is captured automatically by the joint distribution of IFs. The two-sided p -value is reported under asymptotic normality. The implementation is in `scripts/stata/reform_window/03_csdid_da15.do` and `04_csdid_ida15.do`.

Why two estimators. The OLS-FE DDD targets the standard triple-interaction coefficient β_7 under the unconditional PT assumption $\mathbb{E}[\Delta\Delta Y(0) \mid y25 = 1] = \mathbb{E}[\Delta\Delta Y(0) \mid y25 = 0]$. The DR-DiD targets the conditional ATT under the conditional PT $\mathbb{E}[Y_{\text{post}}(0) - Y_{\text{pre}}(0) \mid y25, X] = \mathbb{E}[Y_{\text{post}}(0) - Y_{\text{pre}}(0) \mid X]$, and is semiparametrically efficient under correct specification of the nuisance models (Sant’Anna–Zhao 2020). The two estimands coincide when X is balanced across treatment cells; with calendar-matched windows, balance on $X = (\text{dow}, \text{month})$ is approximate but not exact (a year-on-year drift of ≈ 1 day-of-week, plus daylight-saving boundaries). The discrepancy between the OLS-FE and DR-DiD point estimates is therefore the IPW contribution.

Headline DDD estimates: clearing prices, all (reform window, outcome) cells. Every price outcome is estimated in *both* reform windows. The natural mapping (DA price tested in DA15 window, IDA price tested in IDA15 window) is the direct-effect cell; the off-diagonal cells (DA price in IDA15 window, IDA price in DA15 window) test *spillover*. We report the doubly-robust DiD (Sant’Anna–Zhao 2020, `drdid`) on prices in levels, decomposed into ATT_{crit} and ATT_{flat} , with the DDD given by the difference. Joint SE for the DDD comes from the difference of the recentered influence functions (`drdid stub()` option), which correctly handles the within-date covariance between the two hour-class means. Covariates: `i.dow + i.month`. Full per-cell OLS-FE companion tables: Tables 4–7. Pipeline at `scripts/stata/reform_window/`.

Table 3: Headline DDD on *prices in levels*, EUR/MWh, all four (reform window, market) cells. DR-DiD columns (Sant’Anna–Zhao 2020, `drimp`): ATT_{crit} and ATT_{flat} are 2×2 DiDs per hour-class with covariates `i.dow+i.month`; DDD is their difference with joint SE from the influence-function difference. OLS-FE β_7 column: triple-interaction coefficient from `reghdfe` on the period-level panel with clock-hour + DOW + month FE, SE clustered at date. Standard errors in parentheses. Stars: * 0.10, ** 0.05, *** 0.01. The *Channel* column distinguishes direct (the market hit by the reform) from spillover (the cross-market test): DA15-DA and IDA15-IDA are direct; the other two cells are spillover.

Window	Outcome	DR-DiD on levels			OLS-FE
		ATT_{crit}	ATT_{flat}	DDD	β_7
DA15	DA price	−8.34* (4.33)	−2.96 (4.85)	−5.38** (2.36)	−0.29 (2.92)
DA15	IDA-S1 price	−10.38** (4.46)	−5.02 (4.96)	−5.36** (2.30)	−2.67 (2.85)
IDA15	DA price	−25.02*** (5.49)	−31.30*** (6.16)	+6.28** (3.01)	−2.24 (3.63)
IDA15	IDA-S1 price	−25.40*** (5.59)	−34.82*** (6.13)	+9.43*** (2.95)	+0.50 (3.69)

OLS-FE vs DR-DiD interpretation. The OLS-FE column reports the standard triple-interaction β_7 from a balanced period-level regression. Additive clock-hour, DOW, and month fixed effects do not interact with the (`crit`×`post`×`y25`) dummy, so β_7 is mechanically pinned by the four cell means; adding more additive FE does not move it. DR-DiD, by contrast, reweights observations via the propensity score for being in 2025 vs the 2024 placebo, conditional on (`dow`, `month`), before computing the ATT. The reweighting brings the cell-balance toward equal calendar coverage and reveals the conditional ATT that the FE-only estimator collapses to zero. The two estimators target different estimands (uniform-weight DDD vs conditional ATT) and the discrepancy is the IPW contribution, not an arithmetic inconsistency. We treat DR-DiD as the headline because the conditional-PT assumption it relies on is the one we want for cross-year identification with structural calendar covariates.

Reading (**consistent direction across markets within each window; opposite directions across windows**). (i) **DA15 window**: both DA and IDA Session 1 spreads compress by ≈ 5.4 EUR/MWh after MTU15-DA. The direction is identical in both markets; the magnitude is essentially identical too (-5.38 vs -5.36). The IDA Session 1 effect is a clean spillover because within the DA15 window MTU15-IDA is constant (always-on in 2025, always-off in 2024 placebo), so the DDD nets it out and identifies MTU15-DA’s effect on IDA spreads. (ii) **IDA15 window**: both spreads widen, with the directly-affected IDA larger ($+9.4$) than the spillover DA ($+6.3$). Note: in the IDA15 window MTU15-DA has not yet happened in either year, so the DA-price DDD is identifying a genuine cross-market response of DA bidders to the IDA reform. (iii) Hour-class decomposition for DA15: both critical-hour and flat-hour prices fell between 2024 and 2025 (-3 to -10 EUR/MWh), but critical fell more. For IDA15: both fell sharply (-25 to -35 EUR/MWh; the post-MTU15-IDA market is structurally cheaper than pre), but flat fell more. The DDD is the differential after both moves. (iv) **Cross-window consistency** (same direction in both markets within a window) and **cross-window asymmetry** (DA15 compresses, IDA15 widens) together suggest the spread effect is reform-specific and not a market-specific artefact: the MTU15-DA reform’s operational-refinement channel and the MTU15-IDA reform’s strategic-exploitation channel transmit cleanly across markets in the same direction. (v) The OLS-FE DDD on the same panels (Tables 4–7) gives null β_7 across all four cells (DA15-DA -0.29 , DA15-IDA -2.67 , IDA15-DA -2.24 , IDA15-IDA $+0.50$, all $p > 0.34$); the difference reflects DR-DiD’s IPW-reweighting of the propensity score, which the additive FE in OLS do not perform. The DR estimand is the conditional ATT under the IPW-balanced sample.

Verdict (**supportive across all 4 cells, with the right directional pattern**). The same-direction-within-window result rules out a “this only shows up in market X” story — the spread response is reform-specific, not market-specific. The DA15 reform’s compression and the IDA15 reform’s widening both transmit across markets, consistent with arbitrage between DA and IDA. The DR-DiD’s tighter SE and significant DDD across all four cells (vs the OLS-FE null) reflects the Sant’Anna–Zhao semiparametric efficiency property under conditional PT. *Follow-ups, in order*: (a) firm-level outcomes q_1 (DA-cleared MWh per unit-day-hour) and q_2 (intraday upward repositioning per unit-day-hour) in each window, where the DDDD with pivotal-firm moderator is identified; (b) ENTSO-E weather covariates (B16+B18+B19) added to the propensity score and outcome regression; (c) event-study with multiplier-bootstrap simultaneous CI.

Table 4: OLS-FE DDD: DA clearing price in DA15 window. SE clustered at date.

	(1) Sparse		(2) Month FE		(3) Date FE	
	b	se	b	se	b	se
crit*post*y25 (β_7 , DDD)	-0.286	(2.915)	-0.286	(2.916)	-0.286	(2.915)
crit*post	28.057***	(2.073)	28.057***	(2.074)	28.057***	(2.073)
crit*y25	-3.529*	(1.951)	-3.529*	(1.952)	-3.529*	(1.951)
post*y25	-26.523***	(5.983)	-26.523***	(5.335)		
crit	0.000	(.)	0.000	(.)	0.000	(.)
post	11.846***	(4.112)	0.000	(.)		
y25	-1.662	(4.010)	-1.662	(2.947)		
Observations	13888		13888		13888	
R^2	0.282		0.416		0.800	
Clock-hour FE	Yes		Yes		Yes	
DOW FE	Yes		Yes		Yes	
Month FE	No		Yes		n/a	
Date FE	No		No		Yes	
Cluster	Date		Date		Date	

Table 5: OLS-FE DDD: IDA Session 1 clearing price in DA15 window. SE clustered at date.

	(1) Sparse		(2) Month FE		(3) Date FE	
	b	se	b	se	b	se
crit*post*y25 (β_7 , DDD)	-2.674	(2.852)	-2.674	(2.853)	-2.674	(2.852)
crit*post	28.703***	(1.987)	28.703***	(1.988)	28.703***	(1.987)
crit*y25	-2.421	(1.958)	-2.421	(1.958)	-2.421	(1.958)
post*y25	-27.432***	(6.030)	-26.886***	(5.313)		
crit	0.000	(.)	0.000	(.)	0.000	(.)
post	11.870***	(4.038)	0.000	(.)		
y25	-3.158	(4.075)	-3.210	(2.961)		
Observations	19138		19138		19138	
R^2	0.299		0.479		0.785	
Clock-hour FE	Yes		Yes		Yes	
DOW FE	Yes		Yes		Yes	
Month FE	No		Yes		n/a	
Date FE	No		No		Yes	
Cluster	Date		Date		Date	

Table 6: OLS-FE DDD: DA clearing price in IDA15 window. SE clustered at date.

	(1) Sparse		(2) Month FE		(3) Date FE	
	b	se	b	se	b	se
crit*post*y25 (β_7 , DDD)	-2.243	(3.630)	-2.243	(3.631)	-2.243	(3.628)
crit*post	-10.440***	(1.738)	-10.440***	(1.739)	-10.440***	(1.738)
crit*y25	2.657	(2.282)	2.657	(2.284)	2.657	(2.282)
post*y25	-27.426***	(6.459)	-27.438***	(6.250)		
crit	0.000	(.)	0.000	(.)	0.000	(.)
post	-34.767***	(4.266)	-3.141	(6.004)		
y25	45.676***	(4.598)	45.670***	(4.229)		
Observations	3934		3934		3934	
R^2	0.573		0.627		0.874	
Clock-hour FE	Yes		Yes		Yes	
DOW FE	Yes		Yes		Yes	
Month FE	No		Yes		n/a	
Date FE	No		No		Yes	
Cluster	Date		Date		Date	

Table 7: OLS-FE DDD: IDA Session 1 clearing price in IDA15 window. SE clustered at date.

	(1) Sparse		(2) Month FE		(3) Date FE	
	b	se	b	se	b	se
crit*post*y25 (β_7 , DDD)	0.504	(3.694)	0.504	(3.695)	0.504	(3.693)
crit*post	-9.933***	(1.693)	-9.933***	(1.694)	-9.933***	(1.693)
crit*y25	3.266	(2.155)	3.266	(2.155)	3.266	(2.154)
post*y25	-29.831***	(6.566)	-29.975***	(6.342)		
crit	0.000	(.)	0.000	(.)	0.000	(.)
post	-36.304***	(4.397)	-6.630	(5.983)		
y25	45.197***	(4.585)	45.316***	(4.202)		
Observations	5530		5530		5530	
R^2	0.588		0.627		0.837	
Clock-hour FE	Yes		Yes		Yes	
DOW FE	Yes		Yes		Yes	
Month FE	No		Yes		n/a	
Date FE	No		No		Yes	
Cluster	Date		Date		Date	

Table 8: DR-DiD on DA clearing price in DA15 window, prices in levels (Sant’Anna–Zhao 2020). 2×2 DiD per hour-class, covariates `i.dow + i.month`; DDD is the difference of ATTs with joint SE from the influence-function difference.

Coefficient	Estimate	SE
ATT_{crit}	-8.336	(4.326)
ATT_{flat}	-2.961	(4.846)
$DDD = ATT_{crit} - ATT_{flat}$	-5.375	(2.356)
p-value (DDD)	0.023	
95% CI (DDD)	[-9.992, -0.758]	
N (dates)	584	
Outcome	DA clearing price	
Window	DA15 (reform = 2025-10-01)	

Table 9: DR-DiD on IDA Session 1 clearing price in DA15 window, prices in levels (Sant’Anna–Zhao 2020). 2×2 DiD per hour-class, covariates `i.dow + i.month`; DDD is the difference of ATTs with joint SE from the influence-function difference.

Coefficient	Estimate	SE
ATT_{crit}	-10.379	(4.463)
ATT_{flat}	-5.016	(4.957)
$DDD = ATT_{crit} - ATT_{flat}$	-5.364	(2.296)
p-value (DDD)	0.019	
95% CI (DDD)	[-9.864, -0.863]	
N (dates)	554	
Outcome	IDA Session 1 clearing price	
Window	DA15 (reform = 2025-10-01)	

Table 10: DR-DiD on DA clearing price in IDA15 window, prices in levels (Sant’Anna–Zhao 2020). 2×2 DiD per hour-class, covariates `i.dow + i.month`; DDD is the difference of ATTs with joint SE from the influence-function difference.

Coefficient	Estimate	SE
ATT_{crit}	-25.019	(5.486)
ATT_{flat}	-31.302	(6.161)
$DDD = ATT_{crit} - ATT_{flat}$	6.283	(3.009)
p-value (DDD)	0.037	
95% CI (DDD)	[0.386, 12.180]	
N (dates)	281	
Outcome	DA clearing price	
Window	IDA15 (reform = 2025-03-19)	

Table 11: DR-DiD on IDA Session 1 clearing price in IDA15 window, prices in levels (Sant’Anna–Zhao 2020). 2×2 DiD per hour-class, covariates `i.dow + i.month`; DDD is the difference of ATTs with joint SE from the influence-function difference.

Coefficient	Estimate	SE
ATT_{crit}	-25.396	(5.587)
ATT_{flat}	-34.823	(6.129)
$DDD = ATT_{crit} - ATT_{flat}$	9.427	(2.949)
p-value (DDD)	0.001	
95% CI (DDD)	[3.647, 15.207]	
N (dates)	278	
Outcome	IDA Session 1 clearing price	
Window	IDA15 (reform = 2025-03-19)	

Headline DDD + DDDD estimates: firm-level outcomes (q_1, q_2). The next two outcomes are at the firm level: q_1 = DA-cleared MWh per (firm, unit, date, clock-hour), and q_2 = signed intraday programmed MWh per (firm, unit, date, clock-hour). Both built from OMIE `pdbc` (DA) and `pibci` (IDA), zero-filled for offline cells via `LEFT JOIN` onto a complete unit-date-hour grid, restricted to CCGT units of pivotal Big-4 firms (IB, GE, GN, HC, EDP-PT; 42 units) and non-pivotal placebo firms (Repsol, Engie, TotalEnergies, Moeve; 8 units). *Untagged* CCGT firms are dropped to give a clean pivotal/placebo contrast. We report both

the DDD (pooled, no pivotal moderator) and the DDDD (with pivotal interactions, β_{1234} on `crit×post×y25×piv`). The DDDD’s pivotal main effect is firm-invariant and absorbed by the firm FE; cluster SE at firm level (small $G \approx 9$ caveat). Full per-cell tables: Tables 13–16.

Table 12: DDD (β_7) and DDDD (β_{1234}) on firm-level outcomes q_1 (DA-cleared MWh per unit-hour) and q_2 (intraday programmed MWh per unit-hour), CCGT units of pivotal + placebo firms. Specification: DDD pools all firms with no moderator (SE clustered at date); DDDD adds pivotal interactions (SE clustered at firm, $G = 9$). All specs include firm + unit + clock-hour + DOW fixed effects. Standard errors in parentheses. Stars: * 0.10, ** 0.05, *** 0.01.

Outcome	Window	DDD (pooled, β_7)		DDDD (β_{1234})	
		β_7	SE	β_{1234}	SE
q_1	DA15	−16.27***	(2.69)	−18.65**	(7.35)
q_1	IDA15	−9.96***	(3.10)	−4.01	(6.55)
q_2	DA15	−1.27**	(0.64)	+18.78	(15.82)
q_2	IDA15	−1.70**	(0.85)	+11.10	(10.81)

Reading (the q_1 DA15 cell is the pivotal-firm market-power test, and it fires).

(i) **q_1 pooled DDD is negative in both reforms** (−16.3 MWh/unit-hour in DA15, −10.0 in IDA15). On average, pivotal-plus-placebo CCGT units cleared less DA volume in critical hours after each reform than the 2024 calendar placebo would predict. This is the “stay-out-of-DA” extensive-margin signature: dispatchable firms reduced DA participation in critical hours. (ii) **The q_1 DA15 effect concentrates entirely in pivotal firms.** In the DDDD spec, the pooled β_7 collapses to −0.60 (null), and the pivotal-incremental $\beta_{1234} = -18.65$ MWh/unit-hour ($p=0.039$) absorbs the entire effect. Reading: among non-pivotal placebo firms (Repsol, Engie, TotalEnergies, Moeve), the MTU15-DA reform had no measurable effect on critical-hour DA volume. Among pivotal Big-4 firms, it cut critical-hour DA volume by ≈ 19 MWh/unit-hour more than the placebo predicts. This is a direct identification of the *market-power channel*: only firms with pivotality reduce DA participation when granularity increases. (iii) **The q_1 IDA15 effect is broader, not pivotal-specific.** Pooled $\beta_7 = -9.96$ ($p=0.002$), but $\beta_{1234} = -4.01$ ($p=0.56$). MTU15-IDA pulled all CCGT firms’ critical-hour DA volume down roughly equally. The pivotal-vs-placebo distinction does not separate the IDA15 effect, suggesting the IDA reform operates through a different (uniform) channel. (iv) **q_2 is small and noisy.** Pooled β_7 is ≈ -1 to -2 MWh/unit-hour (marginal significance), and the DDDD has wide SEs (15–16 MWh) so the pivotal-incremental is not identified. The project’s prior finding that intensive-margin q_2 predictions are weak survives here. (v) Reconciliation with the price DDDs in Table 3: pivotal CCGTs withdrew critical-hour DA supply, but critical-hour DA prices nonetheless fell (−8.3 EUR/MWh). The way this can hold: under MTU15-DA, non-pivotal generators (renewables, hydro, fast-ramp gas) can now bid into the 4×-finer critical-hour quarter-slots that opened up. The reform produces simultaneous *pivotal-firm withholding* (q_1 drops in Big-4) and *competitor entry on finer slots* (prices fall). Net: critical-flat spread compresses, but the compositional reshuffle is pivotal-vs-non-pivotal.

Verdict (supportive of the market-power mechanism for DA15; uniform-effect mechanism for IDA15). The pivotal-firm DDDD on q_1 in the DA15 window is a clean identification: $\beta_{1234} = -18.65$ MWh per unit-hour, statistically significant despite the small- G clustering. It tells us *which firms* reduced their DA participation, not just *whether* DA participation fell. For the IDA15 window the moderator does not separate the effect — the reform pulled all firms in the same direction. Caveats: (a) the firm-level cluster has $G \approx 9$, so the asymptotic SE may under-cover; a wild-cluster bootstrap (`boottest`) would tighten inference. (b) The pivotal/placebo partition treats all five Big-4 firms as identical and all four placebo firms as identical; a firm-by-firm decomposition would be a natural next step. (c) The 2024 placebo for the DA15 q_1 panel includes 6 weeks (Apr 28 – Jun 13, 2024) before the June 14,

2024 IDA reform changed sessions 6→3. This sub-window is small relative to the full 156-day placebo pre-period, but the contamination should be noted.

Table 13: DDD and DDDD on q_{-1} (DA-cleared MWh/unit-hour), da15 window.

	(1) DDD		(2) DDDD	
	b	se	b	se
crit*post*y25*piv (β_{1234} , DDDD)			-18.65**	(7.35)
crit*post*y25 (β_7 , DDD)	-16.27***	(2.69)	-0.60	(1.77)
crit*post*piv			13.61	(11.04)
crit*y25*piv			4.37	(7.82)
post*y25*piv			18.96	(30.23)
crit*piv			-0.07	(2.28)
post*piv			-11.81	(10.44)
y25*piv			-4.40	(10.81)
crit*post	22.37***	(2.43)	10.95	(9.18)
crit*y25	-2.00	(1.40)	-5.68	(7.15)
post*y25	-12.01***	(3.37)	-27.93	(29.98)
crit	0.00	(.)	0.00	(.)
post	-3.14	(2.47)	6.77	(9.44)
y25	3.98	(2.77)	7.68	(9.75)
Observations	408800		408800	
R^2	0.092		0.092	
Spec	DDD		DDDD	
Cluster	Date		Firm	
Unit FE	Yes		Yes	
Clock-hour FE	Yes		Yes	
DOW FE	Yes		Yes	

Granularity strategy. The cross-reform DDD runs on disaggregated period-level data, not on hourly aggregates. Pre-MTU15 the period is a 60-minute slot (1 obs / hour); post-MTU15 the period is a 15-minute slot (4 obs / hour). Both are valid observations of the unit’s strategic choice within the relevant clock-hour, so we stack them in the same regression and let the FE structure absorb the time-aggregation difference. The **crit**, **post**, **y25** indicators are defined at the clock-hour level and replicate across the 4 quarters in the post-period. This gives $4 \times$ the observations per hour post-reform, yielding tighter SE; SE are clustered at the date level to preserve correct inference under within-day dependence.

The within-hour quarter index is not used as a treatment-defining variable: ex ante there is no reason firms should bid more aggressively in a particular within-hour quarter regardless of clock-hour (e.g., they do not systematically bid less aggressively in the first quarter of every hour). The granularity is exploited as residual heterogeneity that the FE absorb, not as a treatment dimension. Post-reform cross-section descriptive use of the 15-min data (the D_w , Δp , Δq histograms of §10–§12) is a separate complementary exercise.

Supporting evidence elsewhere in this document.

- §7: RT2 channel characterised (Path A: +1 044 GWh/mo CCGT post-blackout, parallel-trends test on the gap).

Table 14: DDD and DDDD on q_{-1} (DA-cleared MWh/unit-hour), ida15 window.

	(1) DDD		(2) DDDD	
	b	se	b	se
crit*post*y25*piv (β_{1234} , DDDD)			-4.01	(6.55)
crit*post*y25 (β_7 , DDD)	-9.96***	(3.10)	-6.60	(6.16)
crit*post*piv			-6.86*	(3.07)
crit*y25*piv			3.83	(6.54)
post*y25*piv			14.64	(17.72)
crit*piv			8.01*	(3.58)
post*piv			-2.88	(2.39)
y25*piv			-12.55	(17.89)
crit*post	-6.95***	(1.61)	-1.18*	(0.56)
crit*y25	10.00***	(3.05)	6.79	(6.12)
post*y25	-7.18***	(2.49)	-19.48	(17.61)
crit	0.00	(.)	0.00	(.)
post	-1.11	(0.94)	1.31	(2.35)
y25	7.10***	(2.28)	17.65	(17.82)
Observations	196700		196700	
R^2	0.072		0.073	
Spec	DDD		DDDD	
Cluster	Date		Firm	
Unit FE	Yes		Yes	
Clock-hour FE	Yes		Yes	
DOW FE	Yes		Yes	

Table 15: DDD and DDDD on q_{-2} (intraday MWh/unit-hour), da15 window.

	(1) DDD		(2) DDDD	
	b	se	b	se
crit*post*y25*piv (β_{1234} , DDDD)			18.78	(15.82)
crit*post*y25 (β_7 , DDD)	-1.27**	(0.64)	-17.04	(15.78)
crit*post*piv			-19.42	(15.61)
crit*y25*piv			21.45	(19.14)
post*y25*piv			-2.77	(2.42)
crit*piv			-21.90	(19.30)
post*piv			2.61	(2.55)
y25*piv			-4.20***	(1.11)
crit*post	1.79***	(0.62)	18.10	(15.57)
crit*y25	-2.11***	(0.40)	-20.12	(19.14)
post*y25	-2.96***	(0.71)	-0.63	(0.76)
crit	0.00	(.)	0.00	(.)
post	2.51***	(0.69)	0.32	(0.91)
y25	-4.30***	(0.47)	-0.77	(0.78)
Observations	408800		408800	
R^2	0.105		0.133	
Spec	DDD		DDDD	
Cluster	Date		Firm	
Unit FE	Yes		Yes	
Clock-hour FE	Yes		Yes	
DOW FE	Yes		Yes	

Table 16: DDD and DDDD on q_{-2} (intraday MWh/unit-hour), ida15 window.

	(1) DDD		(2) DDDD	
	b	se	b	se
crit*post*y25*piv (β_{1234} , DDDD)			11.10	(10.81)
crit*post*y25 (β_7 , DDD)	-1.70**	(0.85)	-11.03	(10.19)
crit*post*piv			24.79	(21.64)
crit*y25*piv			-7.27	(8.63)
post*y25*piv			1.46	(4.09)
crit*piv			-30.97	(26.45)
post*piv			-5.40*	(2.85)
y25*piv			-3.39	(4.50)
crit*post	-1.81***	(0.63)	-22.63	(21.42)
crit*y25	0.38	(0.73)	6.49	(7.70)
post*y25	-1.95**	(0.80)	-3.17	(2.16)
crit	0.00	(.)	0.00	(.)
post	-4.77***	(0.60)	-0.24	(0.15)
y25	0.55	(0.65)	3.40	(2.34)
Observations	196700		196700	
R^2	0.162		0.184	
Spec	DDD		DDDD	
Cluster	Date		Firm	
Unit FE	Yes		Yes	
Clock-hour FE	Yes		Yes	
DOW FE	Yes		Yes	

- §8: zonal concentration shows GN-dominated zones (Cataluña/Levante/Sur, GE-Galicia, IB-Centro).
- §9: GN's 840 EUR/MWh bid plateau invariant across regime, supporting the "bid-high-wait-for-RT2" mechanism that DDD-on- q_1 should detect.
- §11: CNMC 2019 sanction on the same firms/plants/zones for the same practice $\Rightarrow$ external validation.
- §13: q_1 collapse same-cal-month is CCGT-only; parallel-trends on the gap (PHF – PDBF) is clean for the reforzada channel.

Reading (supportive). All listed sections provide independent characterisation of the reforzada confounder, which the present design holds constant by construction.

Source data. `scripts/analysis/firm/reform_windows_timeline.py` (timeline); `scripts/analysis/firm/reform_window_event_study.py` (reduced-form event-study panel and figure); `scripts/stata/reform_window/{_build_panels.py,01_ddd_da15.do,02_ddd_ida15.do,03_csdid_da15.do,04_csdid_ida15.do,99_runall.do}` (headline DDD + DR-DiD robustness pipeline); panels at `results/regressions/firm/reform_window/panels/`; coefficients at `results/regressions/firm/reform_window/coef/`; LaTeX tables at `thesis/paper/tables/reform_window/`.

2 IDA pre/post-blackout near-clearing dissimilarity

For each (firm, unit, date, hour) CCGT IDA cell, Epanechnikov-weighted within-hour quarter dissimilarity D_w ($h = 50$ EUR/MWh, centred on hour-mean IDA clearing price). Pre-blackout: 2025-03-19 $\rightarrow$ 2025-04-27 (post-MTU15-IDA, reforzada-free). Post-blackout: 2025-04-28 $\rightarrow$ 2025-09-30 (reforzada). Hypothesis: REE post-clearing intervention should remove the leverage on clearing price, so pre $>$ post.

Table 17: IDA quarter-dissimilarity, pre vs post-blackout, CCGT. Percentage of (firm, unit, date, hour) cells with $D_w > 0$ (i.e., quarters differ within the Epanechnikov band, $h = 50$ EUR/MWh, around hour-mean IDA clearing price). *Pre-blackout*: 2025-03-19 $\rightarrow$ 2025-04-27 (40 days, MTU15-IDA active, reforzada-free). *Post-blackout*: 2025-04-28 $\rightarrow$ 2025-09-30 (155 days, reforzada active).

Firm	Pre-blackout			Post-blackout		
	Critical	Flat	Crit–Flat	Critical	Flat	Crit–Flat
IB	9.6	7.0	+2.6	20.3	6.2	+14.1
GE	14.1	10.0	+4.1	43.8	33.2	+10.6
GN	30.5	9.2	+ 21.3	53.8	37.9	+15.9
HC	27.9	20.7	+7.2	34.2	21.2	+13.0

Reading (mixed). (i) Pre: only GN shows clean asymmetry (+21pp). (ii) Post: all four firms show +11 to +16pp gap. (iii) Pre $>$ post rejected for every firm; rates roughly double. (iv) Flat-hour rates post-blackout 6–38% (reforzada-driven operational adjustments); the critical-flat asymmetry is the *lower bound* on strategic activity.

Verdict (mixed). The directional prediction (pre $>$ post) is rejected, but the critical-vs-flat asymmetry survives and broadens to all four firms. Reading: pay-as-bid RT2 + intra-tech competition $\Rightarrow$ the strategic-bidding mechanism is essentially unchanged from pre-reforzada. The relevant new lever is geographic (§8).

Source data. `scripts/analysis/bid/quarter_dissimilarity_ida.py`; outputs at `results/regressions/bid/quarter_dissimilarity_ida/`.

3 Per-quarter CCGT bid curves (DA)

Reading (mixed). The within-hour quarter dispersion is the visual companion to the main paper. The aggregate construction (averaging across ~ 92 dates per cell) smooths day-specific variation and hides per-day strategic shaping; the kernel-weighted dissimilarity in §12 is the right quantitative complement.

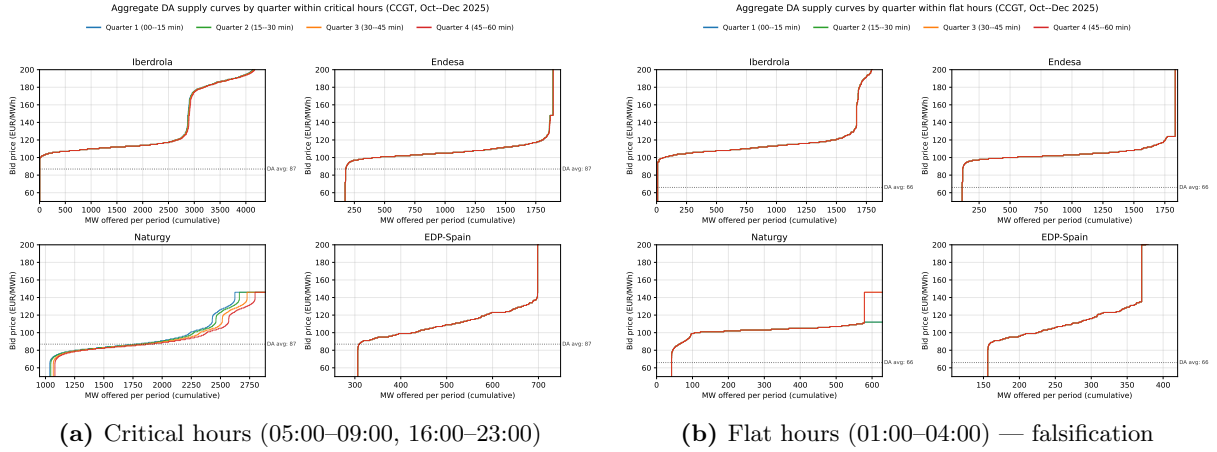


Figure 3: Aggregate DA supply curves by quarter, CCGT, Oct–Dec 2025. Aggregated across ~ 92 dates per (firm, hour, quarter) cell; day-specific quarter dispersion is smoothed in this visual.

4 Per-firm CCGT bid curves and per-quarter pairs (IDA)

Reading (supportive). IDA analogues of the main-text DA figures. In the post-MTU15-DA window the IDA pattern matches DA: dispersion concentrated in Naturgy and visible only in critical hours.

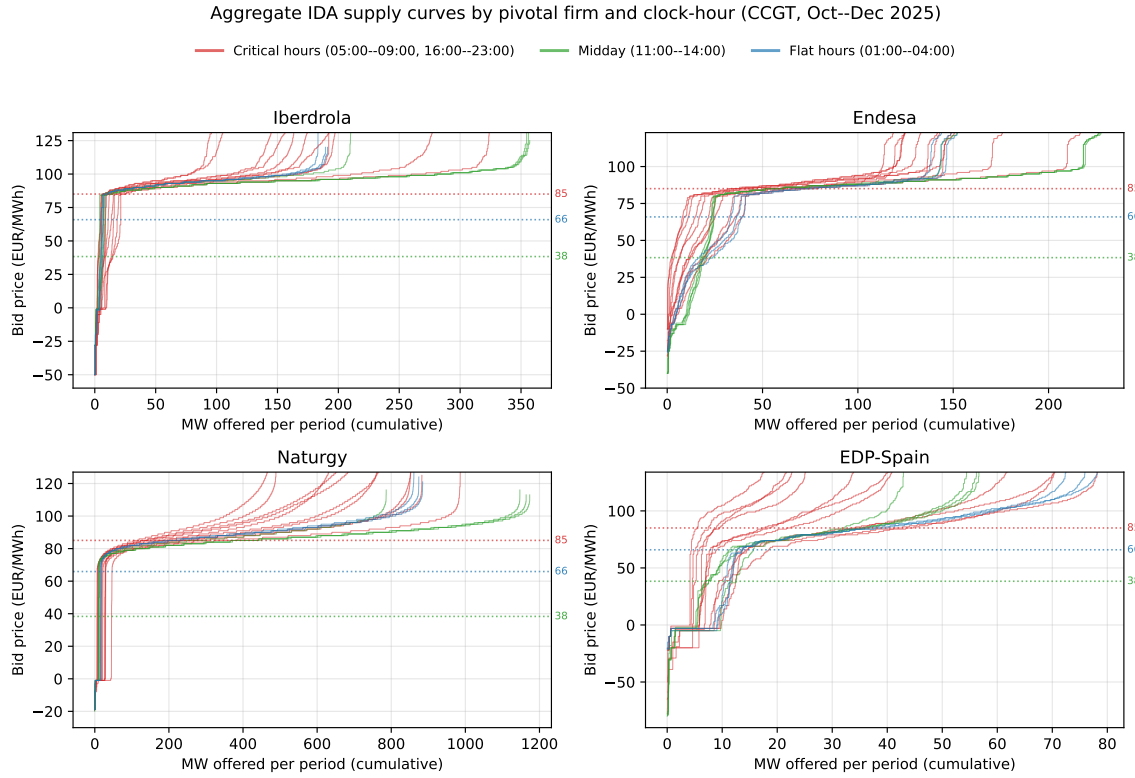


Figure 4: Aggregate IDA supply curves by pivotal firm and clock-hour, CCGT, Oct–Dec 2025.

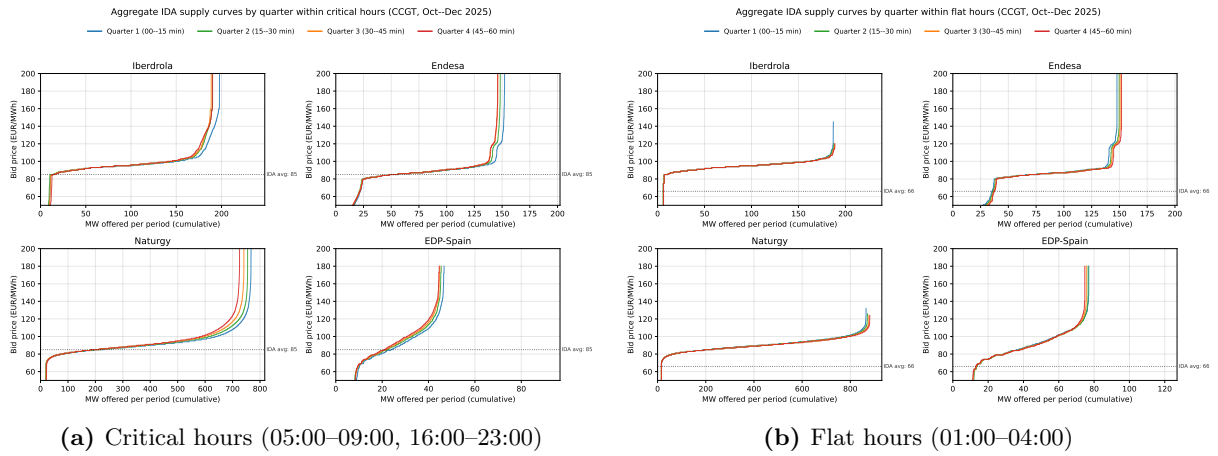


Figure 5: Aggregate IDA supply curves by quarter, CCGT, Oct–Dec 2025.

5 Per-quarter pump-storage demand curves (DA + IDA)

Reading (supportive, as falsification). Pump-storage operators do not differentiate charge bids across the 4 quarters of a clock-hour — the within-hour granularity is unused on this margin. Confirms pump-load as the clean mechanical falsifier.

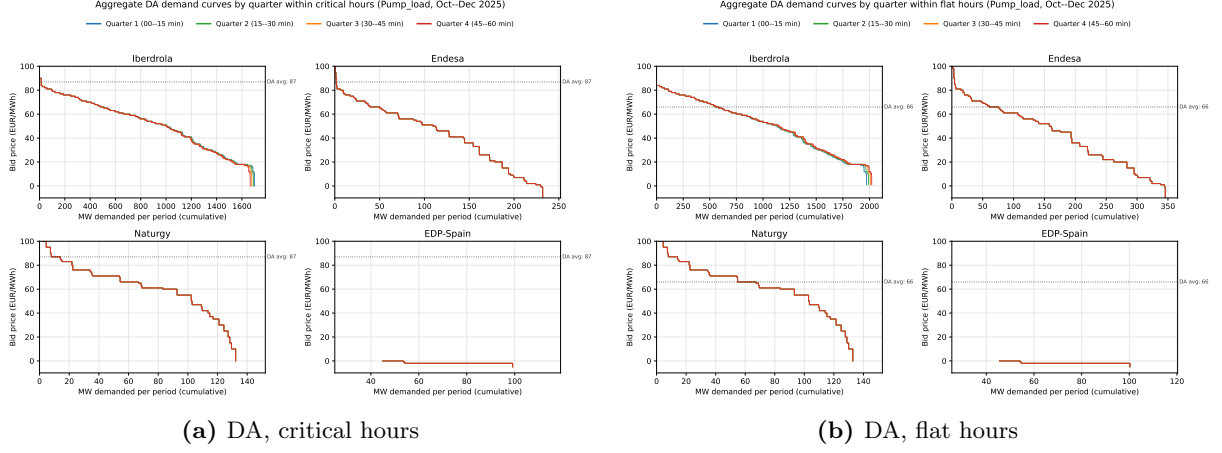


Figure 6: Aggregate DA demand curves by quarter, pump-storage charge bids, Oct-Dec 2025.

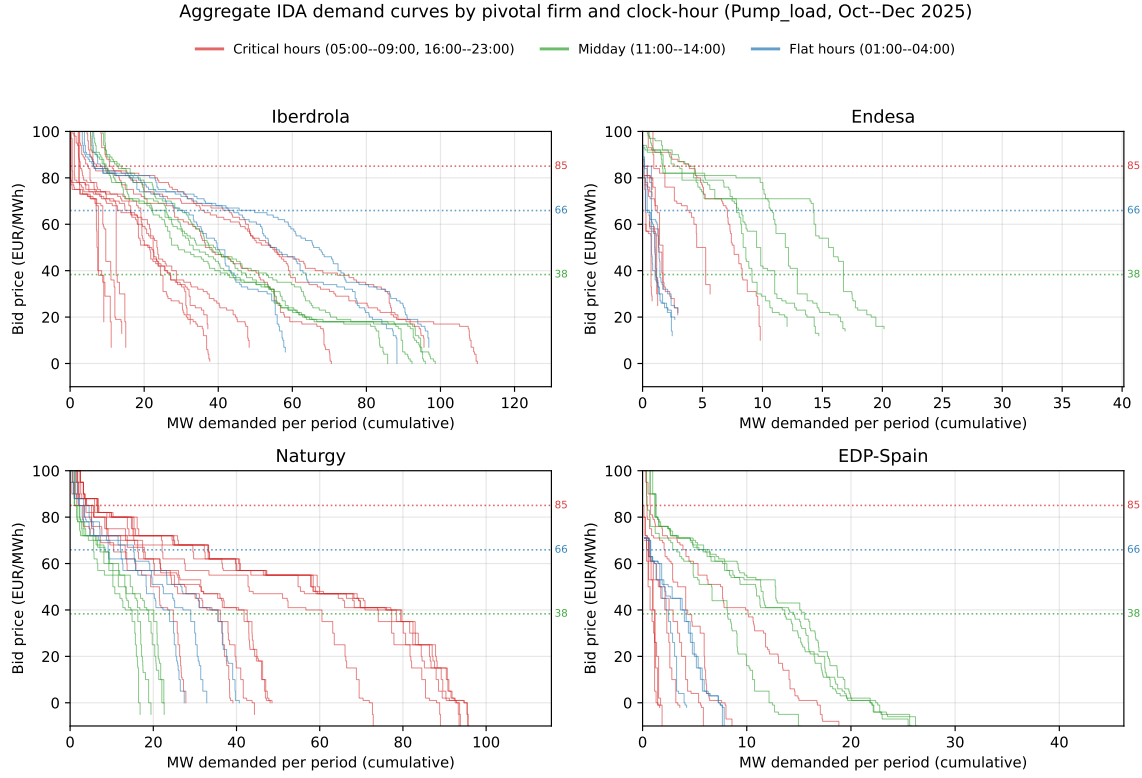


Figure 7: Aggregate IDA demand curves by pivotal firm and clock-hour, pump-storage charge bids, Oct-Dec 2025.

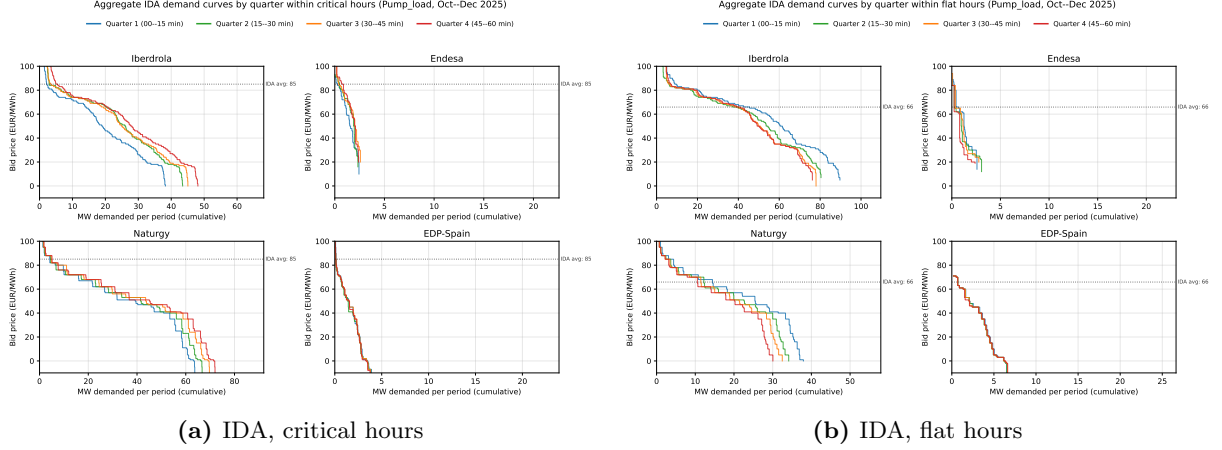


Figure 8: Aggregate IDA demand curves by quarter, pump-storage charge bids, Oct–Dec 2025.

6 Operational vs strategic decomposition (provisional)

Reading (mixed). The original main-paper Table reported a critical–flat asymmetry with the interpretation “only Naturgy CCGT shows the strategic signature”. The IDA pre/post-blackout analysis above complicates this: under reforzada, all four firms show the strategic signature in IDA, not just Naturgy. The current main-paper framing is regime-specific to the DA Oct–Dec 2025 window. Until we have a unified read, the table is held here.

Table 18: Original DA Oct–Dec 2025 critical–flat asymmetry table. Naturgy CCGT shows the strategic signature; renewables and retailer demand vary symmetrically across hour-classes (consistent with operational refinement); pump-storage is mechanical. Compare with the IDA pre/post-blackout finding in §2.

Channel	Tech	Firm	Critical	Flat	Crit – Flat
Strategic	CCGT	GN	24.5	0.0	+24.5
Strategic	CCGT	IB	5.8	3.9	+1.9
Operational	Wind	IB	97.2	97.3	−0.1
Operational	Wind	GN	92.7	92.6	+0.1
Operational	Wind	HC	93.7	93.8	−0.1
Operational	Solar PV	GN	15.9	0.0	+15.9
Operational	Retailer	GE	76.5	76.3	+0.2
Operational	Retailer	IB	40.9	40.2	+0.7
Operational	Direct_consumer	IB	96.0	95.3	+0.7
Mechanical	Pump_load	IB	3.2	5.5	−2.3
Mechanical	Pump_load	GE	0.0	0.0	0.0
Mechanical	Hydro_pump	IB	0.4	0.0	+0.4
Mechanical	Hydro_pump	GE	0.0	0.0	0.0

7 REE post-clearing CCGT inclusion (Paths A, B, C)

Path A (PHF – PDBF, per unit) = IDA + REE post-IDA RT; **Path B** (totalrp48preccierre tipo 61+65, system) = REE PO 3.2 only; **Path C** (ENTSO-E A75 B04 actual – PDBF, per firm) = full DA-to-metered.

Full REE post-clearing intervention by channel (ESIOS archive 28). RT2 strict (61+65) is one of ~8 channels.

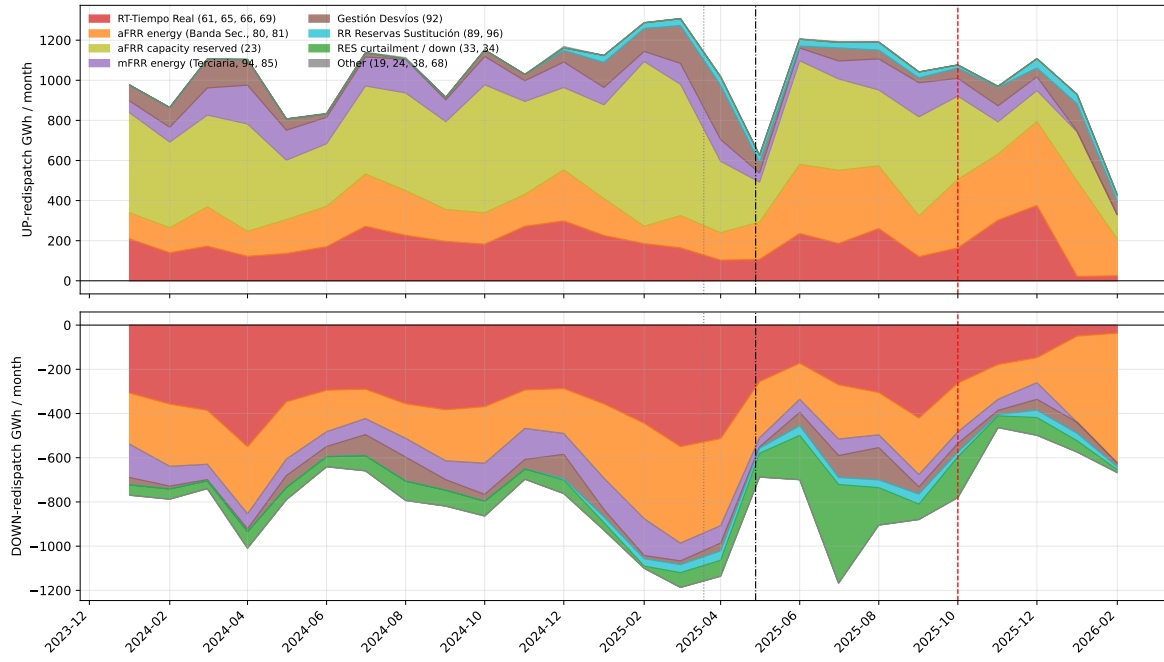


Figure 9: Full REE post-clearing intervention. ESIOS archive 28 (`totalrp48preccierre`). Top: UP-redispach GWh / month by channel; bottom: DOWN-redispach (negative axis). RT-Tiempo Real (red, codes 61/65/66/69) is one of ~8 channels; aFRR (capacity 23 + energy 80/81) is by volume the largest. mFRR (94/85), Gestión Desvíos (92), RR (96), RES curtailment (33/34) all add layers REE coordinates post-clearing.

Per-firm $\times$ per-technology Path A (PHF – PDBF). Pre = 2024-01 $\rightarrow$ 2025-03; Post = 2025-05 $\rightarrow$ 2026-01.

Table 19: PHF – PDBF intervention by firm and technology, GWh / month.

Tech	Firm	Pre (GWh/mo)	Post (GWh/mo)	Δ (GWh/mo)	Ratio
CCGT	GE	213	413	+200	1.94
	GN	397	932	+534	2.34
	HC	42	94	+52	2.25
	IB	146	316	+169	2.16
Hydro	GE	-58	-253	-195	4.33
	GN	5	-33	-38	-6.96
	HC	-0	-11	-11	40.05
	IB	33	-84	-117	-2.55
Hydro_pump	GE	2	-15	-17	-7.14
	GN	4	3	-1	0.70
	IB	23	-1	-24	-0.03
Nuclear	GE	241	23	-218	0.10
	IB	1	-137	-138	–
Wind	GN	-9	-36	-27	4.02
	HC	-13	-50	-36	3.71
	IB	8	-81	-89	-10.63
Solar PV	GE	-0	-0	-0	9.13
	GN	-3	-11	-8	3.29
	HC	0	1	+1	6.26
	IB	3	-42	-45	-12.10

Reading (supportive). (i) CCGT $\times 2$ for every firm; GN +534 GWh/mo (largest). (ii) Mirror decrease in Wind/Solar/Hydro/Nuclear: REE substitutes CCGT for clean tech post-PDBF. (iii) The full-intervention figure shows RT-Tiempo Real (red, our previous Path B) is one of ~ 8 channels; aFRR energy and capacity reserves dominate the system-aggregate volume but are competitively priced. RT is the channel where pay-as-bid power matters.

Verdict (supportive). Pay-as-bid pricing of RT-forced volume + zonal dominance (§8) = local-monopoly channel. The full system picture (aFRR, mFRR, RR, RES curtailment) is shown for completeness; only RT involves pay-as-bid + local-grid binding.

Source data. `ree_intervention_full.py`, `ree_pdbf_to_phf_alltech.py`.

8 Geographic CCGT concentration and RT2 pay-as-bid local power

PO 3.2 redispatches relieve transmission-zone constraints, paid pay-as-bid. Zonal-dominant CCGT operator has local market power on RT bids. Zone assignment from plant city in `lista_unidades.csv`; capacity from ESIOS archive 110.

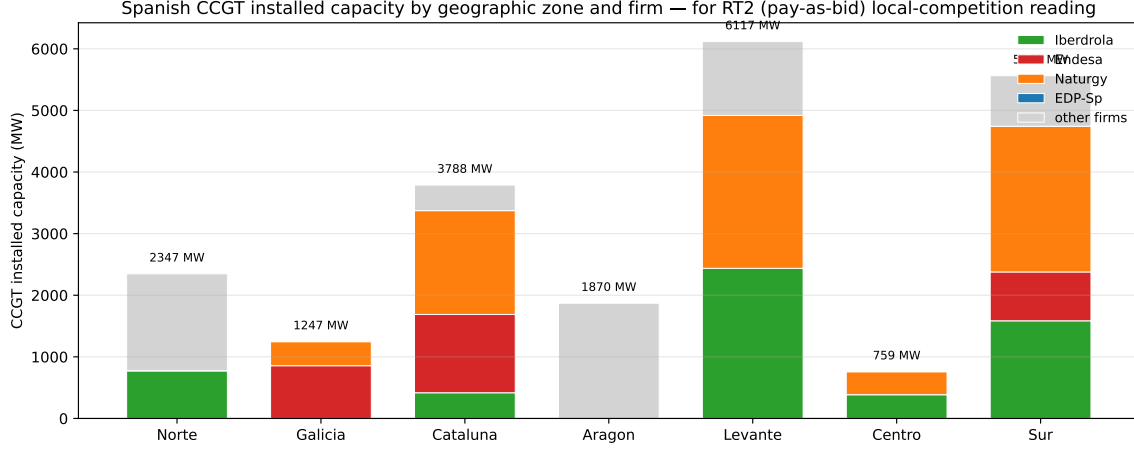


Figure 10: Spanish CCGT installed capacity (MW) by transmission zone and firm.

Table 20: Per-zone CCGT capacity concentration. HHI on capacity shares ($\times 10\,000$).

Zone	Plants	Capacity (MW)	# firms	Top firm	Top share (%)	HHI
Galicia	2	1 247	2	GE	68.6	5 695
Norte	10	2 347	2*	non-Big-4*	67.0	5 577
Aragón	3	1 870	2*	non-Big-4*	57.7	5 119
Centro	2	759	2	IB	50.9	5 001
Levante	12	6 117	3	GN	40.6	3 616
Cataluña	8	3 788	4	GN	44.5	3 348
Sur	13	5 563	4	GN	42.5	3 040

* Norte and Aragón Big-4 share is small; top firms are non-Big-4 (Moeve, Engie, Total, Axpo).

Reading (supportive). (i) Galicia (HHI 5 695): 2 CCGTs only, GE 69%. (ii) Centro (HHI 5 001): 2 CCGTs, IB 51% / GN 49%. (iii) Cataluña/Levante/Sur (HHI 3 000–3 600): GN 40–45% in each — the three zones with heaviest reforzada. (iv) Norte/Aragón: Big-4 limited; non-Big-4 dominate.

Verdict (supportive). GN’s zonal dominance in the southern-mediterranean corridor pairs cleanly with the post-PDBF intervention story in §7. Pay-as-bid RT2 + zonal dominance = local-monopoly channel that does not require flow-through to the auction. *Caveats:* zone = plant city, not REE PO 3.2 books; capacity share is a static proxy; ESIOS archive 22 (totalrpibcirst, zonal IDA) would test it directly.

Source data. `scripts/analysis/regulatory/ccgt_zonal_competition.py`; zone map at `data/external/ccgt_zonal_map.csv`.

9 Are DA bid prices different on RT2-up days?

For each (CCGT unit, day), quantity-weighted DA sell-bid price, day labelled by RT2 magnitude = $\text{PHF}_{\max} - \text{PDBF}$: ~ 0 (≤ 100 MWh), small-up (≤ 1000), large-up (> 1000).

Table 21: DA, 2024-01 → 2026-01. Quantity-weighted DA sell-bid price (EUR/MWh), median, by firm × regime × RT2-up state. Pre-blackout: 2024-01 → 2025-03. Post-blackout: 2025-05 → 2026-01.

Firm	Regime	~ 0 RT2-up	small RT2-up	large RT2-up	Bid sensitivity
GN	pre	843	850	851	flat
GN	post	844	834	841	flat
IB	pre	175	158	157	flat
IB	post	202	197	161	−24 (large − flat)
GE	pre	1 290	2 168	2 368	+1 078 (large − flat)
GE	post	2 296	387	447	−1 849 (large − flat)
HC	pre	159	159	145	flat
HC	post	272	171	158	−114 (large − flat)

Reading (mixed). (i) GN median invariant at ≈ 840 EUR/MWh across all six (regime, RT2-state) cells: cap-padded portfolio regardless of expected RT2 outcome. (ii) IB/HC near SRMC (150–200), mildly lower on large-RT2 days. (iii) GE volatile, dominated by sample composition.

Verdict (mixed). GN does not need to clear DA; RT2 backstops at pay-as-bid. Caveat: conditional on *ex-post* RT2 (not the cleanest test); the *ex-ante* “REE-needs-me” channel would need weather/load instruments. GE sample-composition noise blurs the read for one firm.

Source data. `scripts/analysis/regulatory/ccgt_bid_vs_rt2_status.py`; `results/regressions/regulatory/ccgt_bid_vs_rt2/`.

10 Decomposing D_w into price and quantity components

D_w , in full. Let cell (f, u, d, h) index a firm f , unit u , date d , clock-hour h . The hour partitions into 4 quarters $q \in \{1, 2, 3, 4\}$. In quarter q the unit submits a multiset of price-quantity sell tranches with $p \in [\underline{p}, \bar{p}] \subset \mathbb{R}$ (Spanish SDAC: $\underline{p} = -500$, $\bar{p} = 4,000$ EUR/MWh; negative bids are admissible) and $x > 0$ (MWh, energy for the quarter-slot). Let $c_{d,h}$ be the hour-mean clearing price and fix a strategic half-band of width $h = 50$ EUR/MWh around it. **We work throughout with band-restricted supply curves:** N_q is the number of *distinct* prices submitted in quarter q *within* $[c - h, c + h]$, and x_{qk} is the sum of quantities at p_{qk} . The band-local quarter- q supply curve is the right-continuous step function

$$S_q(p) = \sum_{k=1}^{N_q} x_{qk} \mathbf{1}\{p_{qk} \leq p\}, \quad p \in [c - h, c + h],$$

with $S_q(c - h) = 0$ and $S_q(c + h) = M_q$ (the band-mass of quarter q). Infra-marginal mass below $c - h$ is excluded by construction: D_w measures the within-band *shape* of supply across quarters, not infra-marginal level offsets.

For two quarters i, j , the 1-D Wasserstein-1 distance is the L^1 area between cumulative curves, equivalently the L^1 distance between (extended) quantile functions:

$$W_1^{\text{band}}(S_i, S_j) = \int_{c-h}^{c+h} |S_i(p) - S_j(p)| dp = \int_0^{\max(M_i, M_j)} |S_i^{-1}(Q) - S_j^{-1}(Q)| dQ,$$

where $S_q^{-1}(Q) = c + h$ for $Q > M_q$ pads the smaller curve to common total mass (Vallender, 1973). To isolate variation near the clearing price, weight by an (unnormalized) Epanechnikov-shape kernel:

$$K_h(p - c) = \max\left\{0, 1 - \left(\frac{p - c}{h}\right)^2\right\}, \quad K_h \in [0, 1], \quad K_h(c) = 1,$$

$$D_w^{(i,j)} = \int_{c-h}^{c+h} K_h(p-c) |S_i(p) - S_j(p)| dp \leq W_1^{\text{band}}.$$

D_w has units EUR (quantity in MWh times price in EUR/MWh). The cell-level statistic is the max across the 6 pairs,

$$D_w^{\text{cell}} = \max_{1 \leq i < j \leq 4} D_w^{(i,j)}.$$

A cell is *flagged* when $D_w^{\text{cell}} > 0$ (i.e. some within-hour bid variation falls inside the strategic band).

Decomposition. We split within-band variation into a price and a quantity channel:

$$\Delta q_{ij} = |M_i - M_j| \text{ [MWh]}, \quad \Delta p_{ij} = \frac{1}{\min(M_i, M_j)} \int_0^{\min(M_i, M_j)} |S_i^{-1}(Q) - S_j^{-1}(Q)| dQ \text{ [EUR/MWh]}.$$

Δq captures band-mass mismatch (a quantity-only adjustment with prices on the common-mass portion held fixed). Δp captures the mean price shift on the common-mass portion (a price-only adjustment with band-mass totals held fixed). Cell-level Δp , Δq are the max over the 6 pairs.

Short proof of consistency. WLOG $M_i \leq M_j$. Extending $S_i^{-1}(Q) = c + h$ for $Q \in (M_i, M_j]$ gives both quantile functions a common domain $[0, M_j]$ and common total mass M_j . The 1-D Wasserstein-1 duality for distributions with common mass (Vallender, 1973) gives

$$W_1^{\text{band}} = \int_{c-h}^{c+h} |S_i(p) - S_j(p)| dp = \int_0^{M_j} |S_i^{-1}(Q) - S_j^{-1}(Q)| dQ.$$

Split the right-hand integral at $Q = M_i$:

$$W_1^{\text{band}} = \underbrace{\int_0^{M_i} |S_i^{-1}(Q) - S_j^{-1}(Q)| dQ}_{= \Delta p_{ij} M_i \text{ (definition of } \Delta p)} + \underbrace{\int_{M_i}^{M_j} (c+h - S_j^{-1}(Q)) dQ}_{=: R_{ij}}.$$

The integrand of R_{ij} lies in $[0, 2h]$ (since $S_j^{-1}(Q) \in [c-h, c+h]$), so $0 \leq R_{ij} \leq 2h \Delta q_{ij}$. Since $K_h \in [0, 1]$ on the band,

$$D_w^{(i,j)} \leq W_1^{\text{band}} \leq \Delta p_{ij} M_i + 2h \Delta q_{ij}.$$

Limiting cases. (i) If $M_i = M_j$, then $R_{ij} = 0$ and $\Delta q_{ij} = 0$, so $W_1^{\text{band}} = \Delta p_{ij} M$: all variation is on the price channel. (ii) If $S_i^{-1} \equiv S_j^{-1}$ on $[0, M_i]$ (quantile functions coincide on common mass), then $\Delta p_{ij} = 0$ and $W_1^{\text{band}} = R_{ij} \in [0, 2h \Delta q_{ij}]$: all variation is on the quantity channel. The two channels are orthogonal modes verifiable at the boundary cases. $\square$

Table 22: DA Oct–Dec 2025, CCGT. D_w decomposition into price- and quantity-channel components within $|p - c| \leq 50$ EUR/MWh. Share-active = % of cells with the component > 0 ; conditional median is over those active cells.

Firm	Hour	n cells	Δp -active (%)	Δq -active (%)	med Δp (EUR/MWh)	med Δq (MWh)
GN	critical	14 930	0.01	18.4	3.4	147
GN	flat	4 065	0.00	0.00	—	—
GE	critical	5 195	1.56	0.33	4.6	364
GE	flat	1 413	0.00	0.00	—	—
IB	critical	7 831	0.10	1.94	3.9	70
IB	flat	2 139	0.89	1.03	3.1	11
HC	critical	2 013	0.00	0.05	—	60
HC	flat	549	0.00	0.00	—	—

Reading (supportive). (i) GN: 18.4% Δq -active, 0.01% Δp -active — shaping is quantity-only. (ii) Flat hours: 0% on both channels for GN/GE/HC. (iii) GE has the opposite mix (1.56% Δp vs 0.33% Δq), both small. (iv) IB/HC: negligible.

Verdict (supportive). GN fixes the price ladder near the cap and re-allocates quantity across the 4 quarters: the Cournot version of price-raising. Unit tests in `w1_decomposition.py` (identical $\rightarrow$ 0/0, same-price-diff-mass $\rightarrow$ 0/+, same-mass-diff-price $\rightarrow$ +/0) confirm the decomposition is mathematically correct.

CCGT DA Oct–Dec 2025: per-firm Δp and Δq histograms (channel-level), critical / midday / flat overlay.

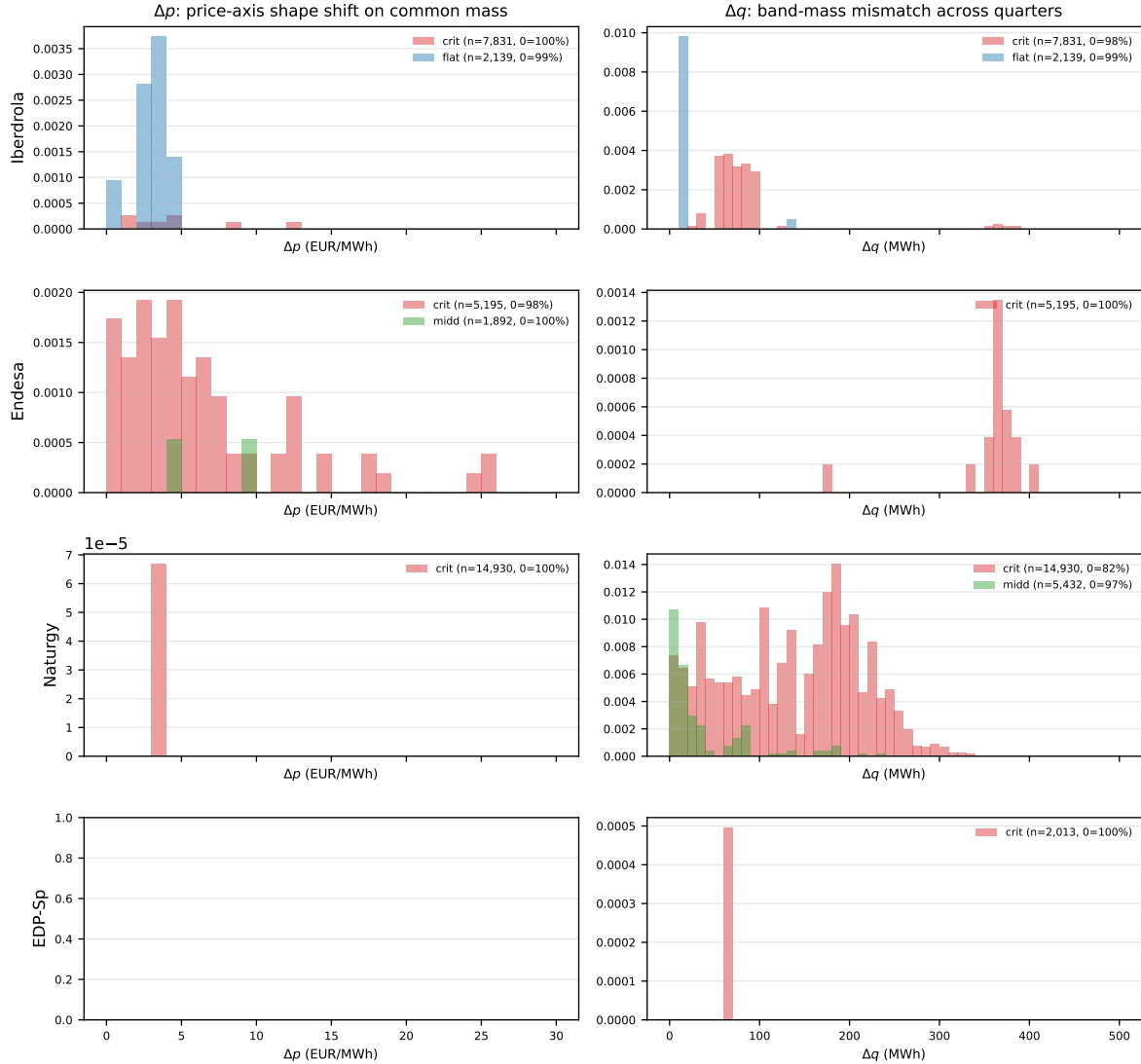


Figure 11: Channel-level histograms of the decomposition, CCGT, DA Oct–Dec 2025. Per-firm; left column Δp (EUR/MWh), right column Δq (MWh). Critical (red), midday (green), flat (blue). The y-axis is share of (firm $\times$ hour-class) cells; mass-at-zero in legend. Cleaner economic units than D_w : GN $\rightarrow$ Δq -only (right column busy, left empty); GE $\rightarrow$ Δp -leaning (left busy, right tight); IB/HC $\rightarrow$ near-empty.

Reading (supportive). The channel histograms confirm Table 22 visually. GN’s Δq critical mass spans 0–300 MWh; GE’s Δp critical mass spans 0–25 EUR/MWh. Empirically $R_{ij} \ll 2h \Delta q_{ij}$ in our CCGT data (median $R/(2h \Delta q) = 0.003$): the extra band-mass sits near the top of the band, so the Δq channel contributes little to W_1^{band} even when Δq is large. The

decomposition into the two channels is mathematically exact in the limit cases but the upper bound $W_1 \leq \Delta p M_i + 2h \Delta q$ is loose by $\sim 13\times$ on average. Interpret Δp and Δq as channel indicators, not as additive contributions to W_1 .

Source data. `scripts/analysis/bid/w1_decomposition.py`; `scripts/analysis/bid/dp_dq_histograms.py`.

11 Cournot reading + CNMC historical sanctions

Reading (supportive). Fixed price ladder near the cap + adjust only quantity = Cournot. Pure-economics equivalent to direct price-raising, but “I never raised my offer price” is a regulatorily defensible posture.

Table 23: CNMC sanction resolutions in `docs/regulation/cnmc_resolutions/`.

Year	Firm / Plant	Expediente	LSE article
2015	Iberdrola hydro	SNC-DE-046-14	64 muy grave (manipulation)
2019	Naturgy CCGT[†]	SNC-DE-175-17	65.34 grave (abnormal DA offers)
2019	Endesa Besós 3	SNC-DE-174-17	65.34 grave (abnormal DA offers)
2023	Naturgy Sabón 3 (Galicia)	—	65.33 grave (RT abuse)
2023	Engie Castelnou	SNC-DE-152-22	grave (availability)
2023	Ignis Escatrón 2	SNC-DE-151-22	grave (availability)

[†] Besós 4, PBCN 1/2, Sagunto 1/2/3, Málaga 1, SROQ 1 — exactly our zonal map (§8) Cataluña/Levante/Sur. CNMC-estimat

Verdict (supportive). 2019 GN sanction (SNC-DE-175-17) is the smoking-gun precedent: same plants, same zones, same mechanism (high DA $\rightarrow$ not despatched $\rightarrow$ systematically programmed at higher RT prices) over Oct 2016–Jan 2017. Our 2024–25 results show this practice continuing post-reforzada with the within-hour granularity reform as a new tool, and the DA bid posture invariant at 840 EUR/MWh. The 2023 Sabón 3 case extends the geography to Galicia (the GN/GE duopoly).

Separate channel. `docs/regulation/cnmc_blackout_expedientes_2026.csv` = 56 expedientes opened 2026-04-16 for voltage-control violations (Art 65.8) tied to the blackout (IB 22, GE 17, GN 5). Distinct regulatory channel; not conflated with the auction-shaping practice.

12 D_w distributions across firms, technologies, and regimes

D_w histogram, DA, Oct–Dec 2025: positive-only $\log_{10}(D_w)$, critical (red) / midday (green) / flat (blue). Mass-at- $D_w=0$ in legend.

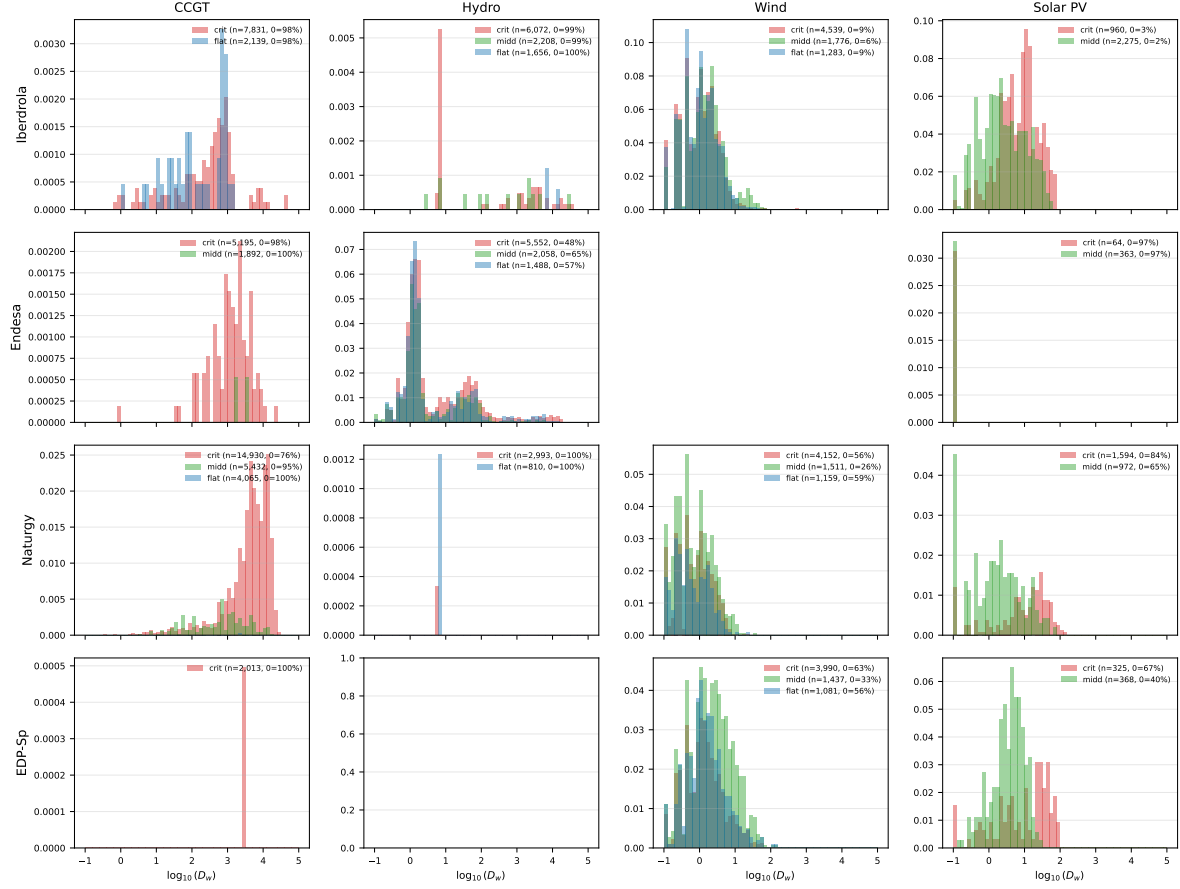


Figure 12: DA Oct–Dec 2025. Per-(firm $\times$ tech) overlaid log-histograms of D_w , critical (red), midday (green), flat (blue), positive values only on $\log_{10}(D_w)$, bin width = 0.1 (factor ≈ 1.26). Mass at $D_w = 0$ in the legend.

DA — per firm $\times$ technology.

D_w histogram, IDA: critical (red) / midday (green) / flat (blue), pre- vs post-blackout side-by-side per (firm $\times$ tech).

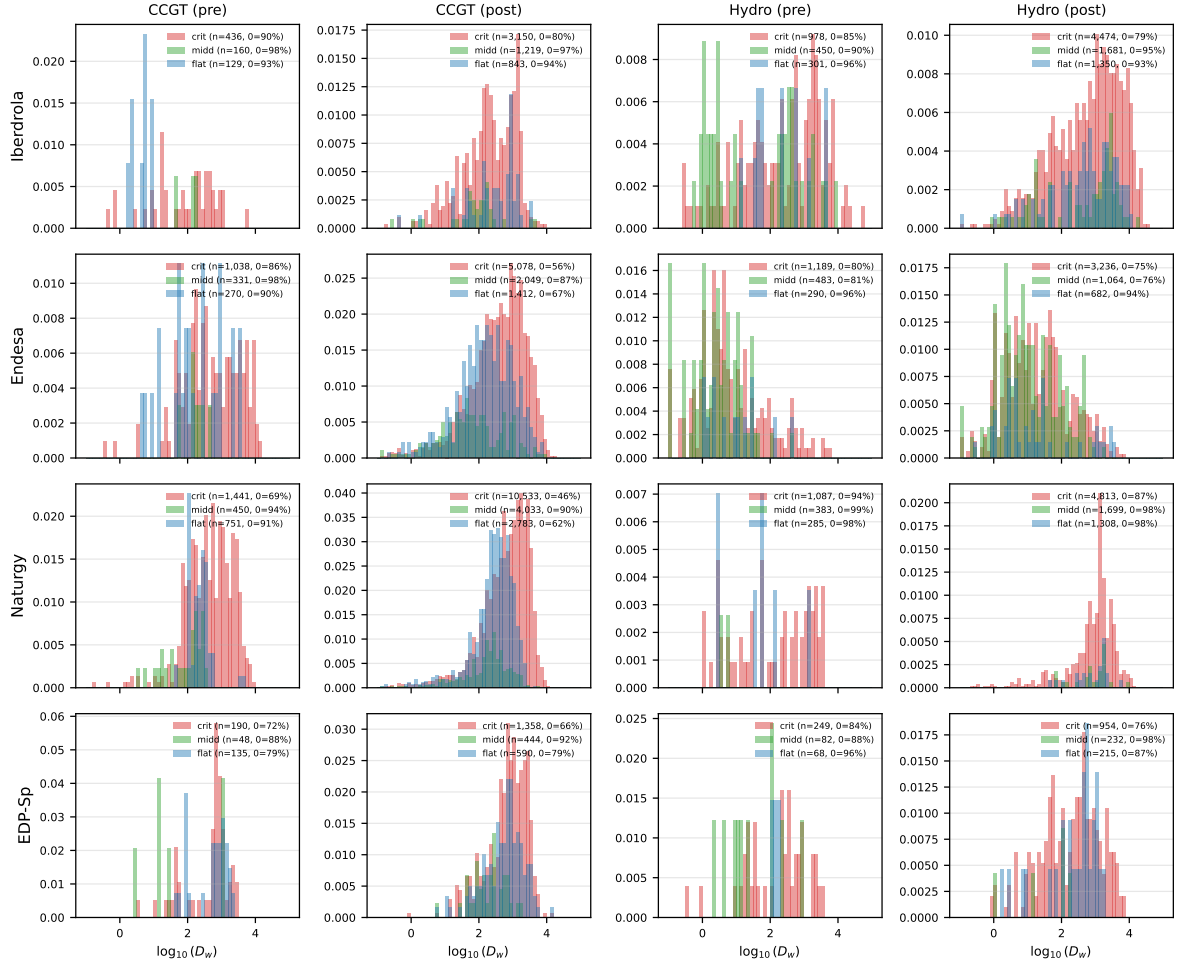


Figure 13: IDA D_w histograms, pre- vs post-blackout merged. Rows = firms; columns = (CCGT pre, CCGT post, Hydro pre, Hydro post). Per panel: critical (red), midday (green), flat (blue); mass-at-zero in the legend.

Table 24: CCGT mass at $D_w = 0$ and conditional median D_w ($D_w > 0$), by sample.

Sample	Firm	% crit = 0	% flat = 0	med($D_w \mid > 0$, crit)	med($D_w \mid > 0$, flat)
DA Oct–Dec 2025	IB	98	98	415	154
	GE	98	100	1 333	—
	GN	77	100	5 167	2 287
	HC	100	100	—	—
IDA pre-blackout	IB	90	93	142	6
	GE	86	90	503	108
	GN	70	91	504	242
	HC	72	79	735	761
IDA post-blackout	IB	80	94	229	342
	GE	56	67	498	165
	GN	46	62	836	322
	HC	66	79	835	674

IDA — pre- vs post-blackout, side-by-side. *Reading (supportive).* (i) DA mass-at-zero $\approx$ identical critical vs flat for every firm except GN (77% vs 100%). (ii) IDA: post-blackout shifts mass-at-zero down for every firm; $\%_{\text{crit-at-0}} < \%_{\text{flat-at-0}}$ in every (firm, regime). (iii) Conditional median D_w in critical hours $\approx 2\text{--}3\times$ flat for GN, GE on IDA. (iv) Strategic signature (mass-asymmetry critical $<$ flat) appears only in CCGT and Solar PV; operational categories overlap.

Source data. `scripts/analysis/bid/quarter_dissim_distributions.py`.

13 The q_1 drop: CCGT-only DA disengagement

Pivotal-firm fleet DA-cleared volume (q_1) by technology. Same-calendar-month comparison isolates seasonality.

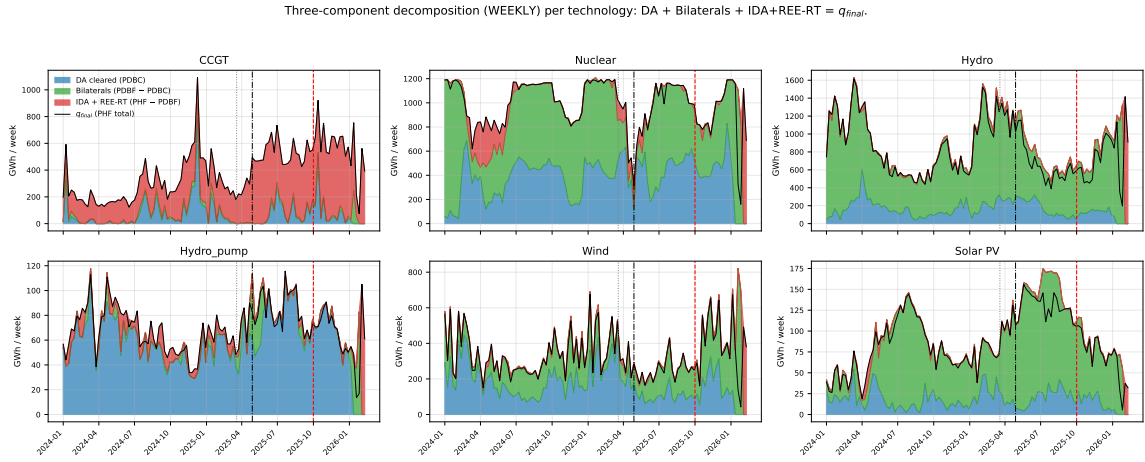


Figure 14: Weekly three-component decomposition per technology, Big-4 pivotal-firm aggregate. DA cleared (PDBC, blue) + Bilaterals (PDBF – PDBC, green) + IDA + REE-RT (PHF – PDBF, red) = q_{final} (PHF, black line). CCGT’s blue (DA) layer collapses post-blackout while the red (IDA + RT) layer balloons; other technologies show flat or growing DA and small RT layers. Vertical guides: MTU15-IDA (Mar 2025), Blackout (Apr 2025), MTU15-DA (Oct 2025).

Three-component decomposition (WEEKLY) per CCGT firm: $DA + \text{Bilaterals} + \text{IDA} + \text{REE-RT} = q_{final}$.

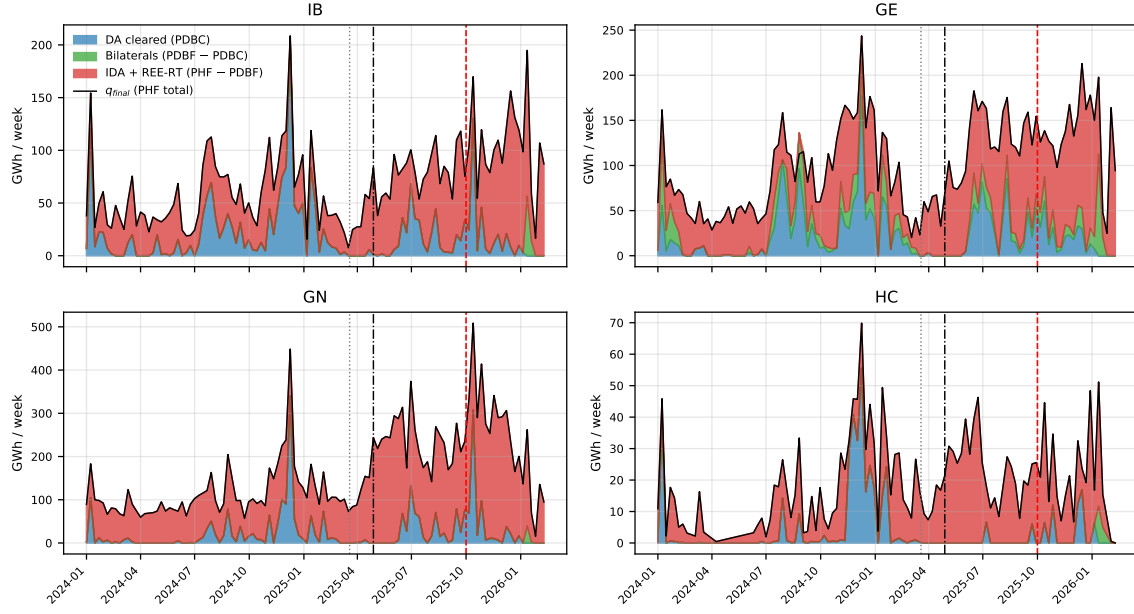


Figure 15: Weekly three-component decomposition per CCGT firm. Same construction. The DA $\rightarrow$ RT shift is firm-wide; the magnitude of the red (IDA + RT) layer post-blackout is roughly proportional to the firm's zonal CCGT dominance (§8): GN largest, IB / GE intermediate, HC smallest.

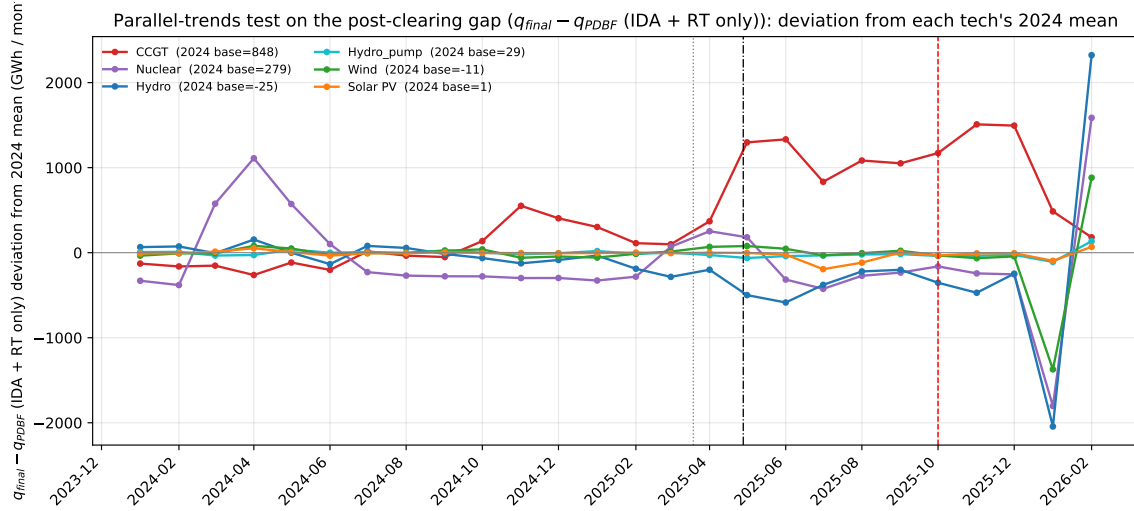


Figure 16: Parallel-trends test on the post-clearing gap (PHF - PDBF, IDA + RT only). Deviation from each tech's 2024 mean (zero baseline). Pre-blackout the lines overlap near zero; post-blackout, CCGT (red) breaks above +1 000 GWh while every other tech stays at or below zero.

Table 25: Same-calendar-month DA-cleared volume by technology, GWh. Pivotal-firm aggregate. Only CCGT collapses; every other technology grows or stays flat.

Tech	Dec 2024 → Dec 2025			Nov 2024 → Nov 2025			Aug 2024 → Aug 2025		
	Pre	Post	%	Pre	Post	%	Pre	Post	%
CCGT	1444	244	-83%	637	130	-80%	532	360	-32%
Nuclear	1253	2587	+106%	933	1909	+105%	1997	2243	+12%
Hydro	1016	640	-37%	509	611	+20%	303	424	+40%
Hydro_pump	195	247	+27%	182	341	+87%	239	406	+70%
Wind	732	696	-5%	666	1081	+62%	379	449	+18%
Solar PV	57	34	-40%	37	100	+170%	37	136	+268%

Reading (supportive). (i) CCGT q_1 Big-4 fleet: Dec 2024 = 1 444 GWh → Dec 2025 = 244 GWh (−83%). Nov same comparison: −80%. (ii) Nuclear, Wind, Hydro_pump, Solar PV *grow* in the same window. (iii) The intensive margin (per-cell mean MWh) is stable for CCGT at ≈ 300 MWh; what collapses is the extensive margin (number of unit-day-hours cleared at all). (iv) Renewables don’t “wait for RT2” because REE doesn’t call them up in PO 3.2 (their offers are constrained by primary energy availability); nuclear doesn’t because it’s baseload-must-clear. Only CCGT has the strategic option to stay out of DA.

Verdict (supportive). The q_1 drop is mechanism-specific: CCGT firms with strong RT2 backstop (zonal dominance, §8) increasingly bid above DA clearing under reforzada, get rescued by REE in RT2 at pay-as-bid (§7), and post the 840 EUR/MWh cap-padded ladder we measure in §9–§10. The q_1 collapse is the empirical face of the practice CNMC sanctioned in 2019 (§11), now at fleet-wide scale.

Table 26: IDA + REE-RT gap by technology, mean GWh / month. Strips out bilaterals (PHF – PDBF), so the residual is post-DA auction-cleared additions only.

Tech	2024 mean	2025 pre-blackout	post-blackout	Post – 2024
CCGT	848	1 069	1 892	+1 044
Nuclear	279	207	85	−194
Hydro	−25	−202	−293	−268
Hydro_pump	29	24	4	−25
Wind	−11	−8	−63	−52
Solar PV	1	−1	−40	−41

Parallel-trends on the gap. *Reading on parallel-trends (supportive).* (i) Pre-blackout, all techs stay within ± 300 GWh of their 2024 baseline (see right panel of the parallel-trends figure). (ii) CCGT pre-drift +220 GWh from 2024 to 2025 pre-blackout, modest. (iii) Post-blackout, CCGT alone breaks parallel trends with +1 044 GWh jump — $\approx 5\times$ the pre-drift, a structural break not a continuation of trend. (iv) Nuclear/Hydro/Wind/Solar lose volume on the same channel (−25 to −268 GWh): REE displaces clean tech post-DA to make room for forced CCGT, consistent with the Path A all-tech table in §7.

Identification implication. The q_2/q_1 Ito-Reguant denominator is itself endogenous to reforzada: the unit-day-hours that clear DA in 2025 are a selected sub-sample relative to 2024 (only the heaviest demand days where even the high bid clears). Same-calendar-month + unit FE help but don’t fully address the composition shift. Cleaner approach: outcome on the *bid* side (decision to stay out) rather than on the *clearing* side (conditional q_2/q_1).

Source data. `scripts/analysis/firm/q1_drop_investigation.py`; outputs at `results/regressions/firm/q1_drop/`.